# NEET 2016 Question Paper with Solutions

## Physics

**Question No. 1**
**Single Correct Answer**
**Topic –** Mechanics
**Concept** – Rotational motion
**Subject Concept** – Moment of inertia
**Concept Field** – Parallel axis theorem
**Question Level** – Easy
**Expected time to solve** – 45 sec.

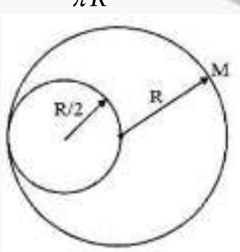
1. From a disc of radius R and mass M, a circular hole of diameter R, whose rim passes through the centre is cut. What is the moment of inertia of the remaining part of the disc about a perpendicular axis, passing through the centre?

(1) 15 $MR^2/32$ (2) 13 $MR^2/32$ (3) 11 $MR^2/32$ (4) 9 $MR^2/32$

**Sol. (2)**

$$I = \frac{MR^2}{2} - \frac{3\sigma}{2}\pi\left(\frac{R}{2}\right)^2\left(\frac{R}{2}\right)^2$$

Where $\sigma = \frac{M}{\pi R^2}$




$$I = \frac{MR^2}{2} - \frac{3}{32}MR^2$$

$$I = \frac{13}{32}MR^2$$

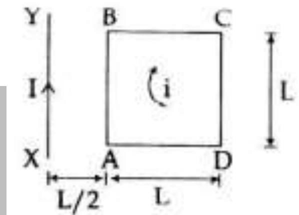
**Question No. 2**
**Single Correct Answer**
**Topic –** Electromagnetism
**Concept** – Magnetic field and magnetic force
**Subject Concept** – Electromagnetic induction

**Concept Field** – Current carrying conductor

**Question Level** - Medium

**Expected time to solve** – 50 sec

2. A square loop ABCD carrying a current i, is placed near and coplanar with a long straight conductor XY carrying a current I, the net force on the loop will be:

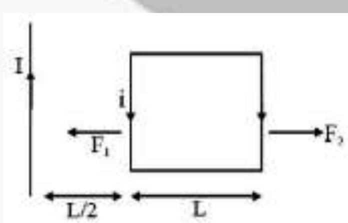



(1) $\dfrac{2\mu_0 Ii}{3\pi}$ (2) $\dfrac{\mu_0\ Ii}{2\pi}$ (3) $\dfrac{2\mu_0\ IiL}{3\pi}$ (4) $\dfrac{\mu_0\ IiL}{2\pi}$

**Sol.** **(1)**

$$F_1 = \frac{\mu_0 IiL}{2\pi \frac{L}{2}} = \frac{\mu_0 Ii}{\pi}$$

$$2 = \frac{\mu_0 IiL}{2\pi \frac{3L}{2}} = \frac{\mu_0 Ii}{3\pi}$$




$$\therefore F_{net} = F_1 - F_2$$

$$F_{net} = \frac{2}{3}\frac{\mu_0 Ii}{\pi}$$

**Question No. 3**

**Single Correct Answer**

**Topic –** Electromagnetism

**Concept** – Magnetic field and magntic force

**Subject Concept** – Magnetic substances

**Concept Field** – Magnetic suscepitibility

**Question Level** - Medium

**Expected time to solve** – 45 sec

3. The magnetic susceptibility is negative for:

(1) diamagnetic material only
(2) paramagnetic material only
(3) ferromagnetic material only
(4) paramagnetic and ferromagnetic materials

**Sol. (1)**

Magnetic susceptibility $\chi m$

is negative for diamagnetic substance only

**Question No. 4**

**Single Correct Answer**

**Topic -** Waves

**Concept** – Sound waves

**Subject Concept** – Frequency of sound

**Concept Field** – Velocity of sound in air

**Question Level** - Medium

**Expected time to solve** – 30 sec

4. A siren emitting a sound of frequency 800 Hz moves away from an observer towards a cliff at a speed of 15 $ms^{-1}$. Then, the frequency of sound that the observer hears in the echo reflected from the cliff is:

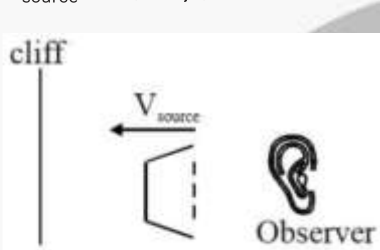
(Take velocity of sound in air = 330 $ms^{-1}$)

(1) 765 Hz (2) 800 Hz (3) 838 Hz (4) 885 Hz

**Sol. (3)**

$f_0$ = 800Hz

$V_{source}$ = 15 m/s




$$f_a = \frac{330}{(330-15)}800 = \frac{330}{315}\times 800$$

$$f_a = 838\,Hz$$

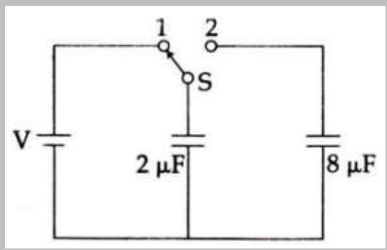
**Question No. 5**

**Single Correct Answer**

**Topic –** Electromagnetism

**Concept** – Current electricity
**Subject Concept** – Electric circuit
**Concept Field** – Stored energy
**Question Level** - Medium
**Expected time to solve** - 45

5.



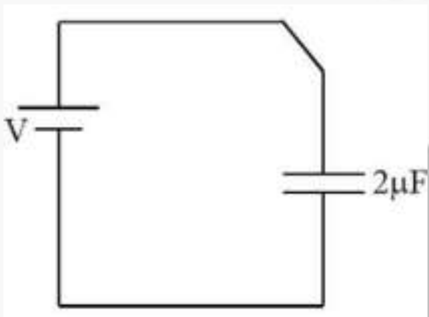

A capacitor of 2μF is charged as shown in the diagram. When the switch S is turned to position 2, the percentage of its stored energy dissipated is:

(1) 0% (2) 20% (3) 75% (4) 80%

**Sol.** **(4)**

Q = 2V

$$U_i = \frac{1}{2} \times \frac{(2V)^2}{2} = V^2$$




$$\therefore V_y = \frac{1}{2} \frac{64V^2}{25 \times 8}$$

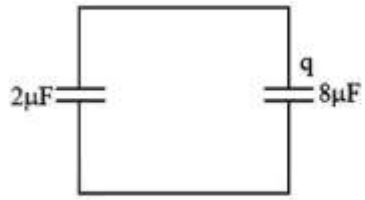
$$\frac{2V - q}{2} = \frac{q}{8} + \frac{1}{2} \frac{4V^2}{25 \times 2}$$

∴ 8V– 4q = q

$$U_f = \frac{5V^2}{25} = \frac{V^2}{5}$$

$$\therefore q = \frac{8V}{5}$$

Energy dissipated $= \frac{4V^2}{5}$




∴ % energy

Dissipated = $\frac{4V^2}{5V^2} \times 100$

= 80%

**Question No. 6**

**Single Correct Answer**

**Topic –** Optics

**Concept** – Wave optics

**Subject Concept** – Diffraction

**Concept Field** – Wavelength

**Question Level** - Hard

**Expected time to solve** -60 sec

6. In a diffraction pattern due to a single slit of width ‘a’, the first minimum is observed at an angle 30° when light of wavelength 5000 Å is incident on the slit. The first secondary maximum is observed at an angle of:

(1) $\sin^{-1}\left(\frac{1}{4}\right)$ (2) $\sin^{-1}\left(\frac{2}{3}\right)$ (3) $\sin^{-1}\left(\frac{1}{2}\right)$ (4) $\sin^{-1}\left(\frac{3}{4}\right)$

**Sol.** **(4)**

a sin 30 = λ

a sin θ $= \frac{3\lambda}{2}$

$$\frac{\sin\theta}{\sin 30} = \frac{3}{2}$$

$$\sin\theta = \frac{3}{2} \times \frac{1}{2}$$

$$\sin\theta = \frac{3}{4}$$

$$\theta = \sin\left(\frac{3}{4}\right)$$

**Question No. 7**

**Single Correct Answer**

**Topic –** Gravitation

**Concept** – Gravitation field

**Subject Concept** – Gravitational potential

**Concept Field** – Height from earth

**Question Level** - Hard

**Expected time to solve** -60 sec

7. At what height from the surface of earth the gravitation potential and the value of g are $-5.4 \times 10^7$J kg$^{-2}$ and 6.0 ms$^{-2}$ respectively? Take the radius of earth as 6400 km:
   (1) 2600 km (2) 1600 km (3) 1400 km (4) 2000 km

**Sol. (1)**

$$V = \frac{GM}{R+h} = -5.4\times10^7$$

$$g = \frac{GM}{(R+h)^2} = 6$$

$$\therefore \frac{5.4}{6}\times10^7 = R+h$$

∴ a × 106 = 6.4 × 106 + h

∴ h = 2600 km

**Question No. 8**

**Single Correct Answer**

**Topic –** Electromagnetism

**Concept** – EM wave

**Subject Concept** – Propagation of EM wave

**Concept Field** – Acceleration charge

**Question Level** – Medium

**Expected time to solve** – 45 sec

8. Out of the following options which one can be used to produce a propagating electromagnetic wave?
   (1) A charge moving at constant velocity
   (2) A stationary charge
   (3) A chargeless particle
   (4) An accelerating charge

**Sol. (4)**

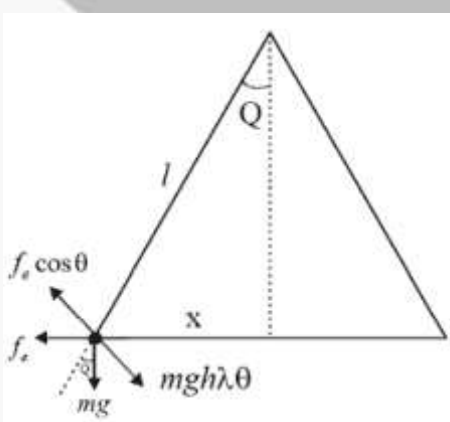
An accelerating charge can produce electromagnetic wave.

**Question No. 9**
**Single Correct Answer**
**Topic –** Electrostatics
**Concept** – Electric charge
**Subject Concept** – Electric force
**Concept Field** – charge distribution
**Question Level** - Hard
**Expected time to solve** – 60 sec

9. Two identical charged spheres suspended from a common point by two massless strings of lengths $l$, are initially at a distance d (d < < $l$) apart because of their mutual repulsion. The charges begin to leak from both the spheres at a constant rate. As a result, the spheres approach each other with a velocity υ. Then υ varies as a function of the distance $x$ between the spheres, as

(1) $v \propto x^{\frac{1}{2}}$ (2) $v \propto x$ (3) $v \propto x^{-\frac{1}{2}}$ (4) $v \propto x^{-1}$

**Sol.** **(3)**




$$\theta = \frac{x}{2l}$$

$f_e \cos\theta = mgh\lambda\theta$

$$f_e = mg \cdot \left(\frac{x}{2}\right)$$

$$\frac{kq^2}{x^2} = \frac{mgx}{2e}$$

$$kq^2 = \frac{mg}{2l} x^3$$

$$\frac{dq}{dt} \propto \frac{3}{2} x^{\frac{1}{2}} \cdot \frac{dx}{dt}$$

$$\Rightarrow x^{\frac{1}{2}} \cdot v = \text{constant}$$

$$v \propto x^{-\frac{1}{2}}$$

**Question No. 10**

**Single Correct Answer**

**Topic –** Waves

**Concept** – Transverse wave

**Subject Concept** – Wavelength

**Concept Field** –

**Question Level** - Medium

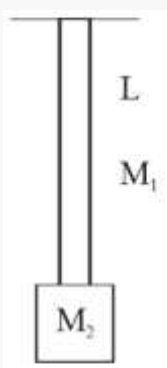
**Expected time to solve** – 45 sec

10. A uniform rope of length L and mass m1, hangs vertically from a rigid support. A block of mass $m_2$ is attached to the free end of the rope. A transverse pulse of wavelength $\lambda_1$ is produced at the lower end of the rope. The wavelength of the pulse when it reaches the top of the rope is $\lambda_2$. The ratio $\lambda_2/\lambda_1$ is:

(1) $\sqrt{\frac{m_1}{m_2}}$ (2) $\sqrt{\frac{m_1+m_2}{m_2}}$ (3) $\sqrt{\frac{m_1}{m_2}}$ (4) $\sqrt{\frac{m_1+m_2}{m_1}}$

**Sol.** **(2)**

At bottom




$$v_1 = \sqrt{\frac{M_2 gL}{M_1}}$$

$$\therefore \lambda_1 = \sqrt{\frac{M_2}{M_1} gL}\frac{1}{f}$$

At top.

$$\therefore \frac{\lambda_2}{\lambda_1} = \sqrt{\frac{M_1+M_2}{M_2}}$$

$$v_1 = \sqrt{\frac{(M_1 + M_2)gL}{M_1}}$$

$$\therefore \lambda_2 = \sqrt{\frac{(M_1 + M_2)gL}{M_1}}\frac{1}{f}$$

**Question No. 11**

**Single Correct Answer**

**Topic –** Thermal physics

**Concept** – Kinetic theory of gases

**Subject Concept** – Refrigreator

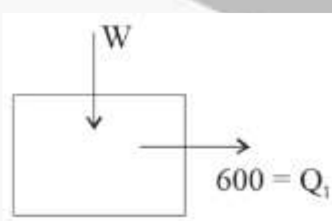
**Concept Field** – Power

**Question Level** - Easy

**Expected time to solve** -50 sec

11. A refrigerator works between 4°C and 30°C. It is required to remove 600 calories of heat every second in order to keep the temperature of the refrigerated space constant. The power required is:
(Take 1 cal = 4.2 Joules)
(1) 2.365 W (2) 23.65 W (3) 236.5 W (4) 2365 W

**Sol.** **(3)**




$$\frac{600 + w}{600} = \frac{303}{277}$$

$$1 + \frac{w}{600} = 1 + \frac{26}{277}$$

$$w = 600 \times \frac{26}{277} \times 4.2$$

w = 236.5

**Question No. 12**
**Single Correct Answer**
**Topic –** Waves
**Concept** – Sound waves
**Subject Concept** – Tunning fork
**Concept Field** – Resonance
**Question Level** - Medium
**Expected time to solve** – 45 sec

12. An air column, closed at one end and open at the other, resonates with a tuning fork when the smallest length of the column is 50 cm. The next larger length of the column resonating with the same tuning fork is:

(1) 66.7 cm (2) 100 cm (3) 150cm (4) 200 cm

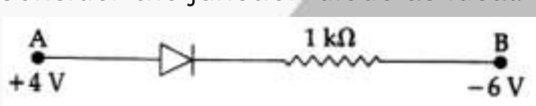
**Sol.** **(3)**

First minimum resonating length for closed organ pipe $\frac{\lambda}{4} = 50\ cm$

$\therefore$ Next larger length of air column = $\frac{3\lambda}{4} = 150\ cm$

**Question No. 13**
**Single Correct Answer**
**Topic –** Modern physics
**Concept** – Semiconductor
**Subject Concept** – Diode
**Concept Field** – p-n Junction
**Question Level** - Hard
**Expected time to solve** -60 sec

13. Consider the junction diode as ideal. The value of current flowing through AB is:




(1) 0 A (2) $10^{-2}\ A$ (3) $10^{-1}\ A$ (4) $10^{-3}\ A$

**Sol.** **(2)**

$V_A - V_3 = 4 - (-6) = 10$

$\therefore i = \frac{10}{1000} = 10^{-2}\ A$

**Question No. 14**
**Single Correct Answer**
**Topic –** Electrostatics
**Concept** – Electric current
**Subject Concept** – Variable resistance
**Concept Field** – Heat
**Question Level** - Medium
**Expected time to solve** – 45 sec

14. The charge flowing through a resistance R varies with time t as Q = at − bt2 where a and b are positive constants. The total heat produced in R is:

(1) $\frac{a^3R}{6b}$ (2) $\frac{a^3R}{3b}$ (3) $\frac{a^3R}{2b}$ (4) $\frac{a^3R}{b}$

**Sol.** **(1)**

$Q = at - bt^2$

$\therefore t \in \left[0, \frac{a}{b}\right]$

$i = \frac{dq}{dt} = a - 2bt$

Note: i is +ve t $\in \left(0, \frac{a}{2b}\right)$

And i is – ve t $\in \left(\frac{a}{2b}, \frac{a}{b}\right)$

Positive current means current one direction and negative current means current in opposite direction.

$\therefore dH = i^2Rdt$

$= (a - 2bt)\ 2R\ dt$

$$H = \int_0^{\frac{a}{b}} (a - 2bt)^2 R dt$$

$$= \left.\frac{(a-2bt)^3 R}{3(-2b)}\right|_0^{\frac{a}{b}}$$

$$= \frac{1}{-b}\left[\left(a - 2b\frac{a}{b}\right)^3 - (a)^3\right] R$$

$$= -\frac{1}{6b}\left[(-a)^3 - a^3\right] R$$

$$H = \frac{a^3R}{3b}$$

**Question No. 15**

**Single Correct Answer**

**Topic –** Thermal physics

**Concept** – Mode of heat transfer

**Subject Concept** – Black body

**Concept Field** – Energy of radiation

**Question Level** – Hard

**Expected time to solve** – 70 sec

15. A black body is at a temperature of 5760 K. The energy of radiation emitted by the body at wavelength 250 nm is $U_1$, at wavelength 500 nm is $U_2$ and that at 1000 nm is $U_3$. Wien's constant, b = 2.88 × $10^6$ nmK.
Which of the following is correct?
(1) $U_1 = 0$ (2) $U_3 = 0$ (3) $U_1 > U_2$ (4) $U_2 > U_1$

**Sol.** **(3)**

$$\lambda_{\min} T = b$$

$$\lambda \propto \frac{1}{T}$$

$$u \propto (T)^4 \propto \frac{1}{(\lambda)^4}$$

So

$$u_1 > u_2$$

**Question No. 16**

**Single Correct Answer**

**Topic –** Solid and fluid mecahnics

**Concept** – Solids

**Subject Concept** – Elasticity

**Concept Field** – Coefficient of linear expansion

**Question Level** - Medium

**Expected time to solve** – 45 sec

16. Coefficient of linear expansion of brass and steel rods are $\alpha_1$ and $\alpha_2$. Lengths of brass and steel rods are $l_1$ and $l_2$ respectively. If $(l_2 - l_1)$ is maintained same at all temperatures, which one of the following relations holds good?
(1) $\alpha_1^2 l_2 = \alpha_2^2 l_1$ (2) $\alpha_1 l_2^2 = \alpha_2 l_1^2$ (3) $\alpha_1^2 l_2 = \alpha_2^2 l_1$ (4) $\alpha_1 l_1 = \alpha_2 l_2$

**Sol.** **(4)**

Difference in length are same so increase in length are equal

$$\Delta l_1 = \Delta l_2$$

$$l_1\alpha_2\Delta T = l_2\alpha_2\Delta T$$

$$\Rightarrow l_1\alpha_1 = l_2\alpha_2$$

**Question No. 17**

**Single Correct Answer**

**Topic –** Modern physics

**Concept** – Semiconductor

**Subject Concept** – Transistor

**Concept Field** – CE configuration

**Question Level** – Medium

**Expected time to solve** – 45 sec

17. A npn transistor is connected in common emitter configuration in a given amplifier. A load resistance of 800 Ω is connected in the collector circuit and the voltage drop across it is 0.8 V. If the current amplification factor is 0.96 and the input resistance of the circuit is 192 Ω, the voltage gain and the power gain of the amplifier will respectively be:

(1) $4, 3.84$ (2) $3.69, 3.84$ (3) 4, 4 (4) 4, 3.69

**Sol.** **(1)**

Voltage gain = $\beta \cdot \left(\frac{R_C}{R_B}\right)$

V = 0.96 $\left(\frac{80}{192}\right)$

V = $\frac{96\times 8}{192} = 4$

And power gain of the amplifier is

$\beta_{ac} \cdot \Lambda_V$

= 0.96 × 4

= 3.84

**Question No. 18**

**Single Correct Answer**

**Topic –** Optics

**Concept** – Wave optics

**Subject Concept** – YDSE

**Concept Field** – Path difference

**Question Level** – Medium

**Expected time to solve** – 45 sec

18. The intensity at the maximum in a Young's double slit experiment is $I_0$. Distance between two slits is d = 5λ, where λ is the wavelength of light used in the experiment. What will be the intensity in front of one of the slits on the screen placed at a distance D = 10 d?

(1) $I_0$ (2) $\frac{I_0}{4}$ (3) $\frac{3}{4}I_0$ (4) $\frac{I_0}{2}$

**Sol.** **(4)**

In YDSE $I_{max} = I_0$

Path difference at a point in front of one of shifts is

$$\Delta x = d\left(\frac{y}{D}\right) = d\left(\frac{\frac{d}{2}}{D}\right) = \frac{d^2}{2D}$$

$$\Delta x = \frac{d^2}{2(10d)} = \frac{d}{20} = \frac{5\lambda}{20} = \frac{\lambda}{4}$$

Path difference is

$$\phi = \frac{2\pi}{\lambda} = (\Delta x) = \frac{2\pi}{\lambda}\left(\frac{\lambda}{4}\right)$$

$$\phi = \frac{\pi}{2}$$

So intensity at that pt is

$$I = I_{max}\cos^2\left(\frac{\theta}{2}\right)$$

$$I = I_0\cos^2\left(\frac{\pi}{4}\right) = \frac{I_0}{2}$$

**Question No. 19**

**Single Correct Answer**

**Topic –** Mechanics

**Concept** – Laws of motion

**Subject Concept** – Circular motion

**Concept Field** – Acceleration

**Question Level** - Easy

**Expected time to solve** -40 sec

19. A uniform circular disc of radius 50 cm at rest is free to tum about an axis which is perpendicular to its plane and passes through its centre. It is subjected to a torque

which produces a constant angular acceleration of 2.0 rad $s^{-2}$. Its net acceleration in $ms^{-2}$ at the end of 2.0 s is approximately: is

(1) 8.0 (2) 7.0 (3) 6.0 (4) 3.0

**Sol.** **(1)**

At the end of 2 sec, $w = w_0 + \alpha t$

$w = 0 + 2(2) = 4$ rad/sec

Particle acceleration towards the center is $= a_c = rw^2$

$a_r = \frac{1}{2}(4)^2$ = 8 m/s

**Question No. 20**

**Single Correct Answer**

**Topic –** Dual nature of radiation and matter

**Concept** – de-broglie wavelength

**Subject Concept** – Photon

**Concept Field** –

**Question Level** - Easy

**Expected time to solve** -30 sec

20. An electron of mass m and a photon have same energy E. the ratio of de-Broglie wavelengths associated with them is:

SS

(1) $\frac{1}{c}\left(\frac{E}{2m}\right)^{\frac{1}{2}}$ (2) $\left(\frac{E}{2m}\right)^{\frac{1}{2}}$ (3) $c(2mE)^{\frac{1}{2}}$ (4) $\frac{1}{c}\left(\frac{2m}{E}\right)^{\frac{1}{2}}$

(c being velocity of light)

**Sol.** **(1)**

De-Broglie wavelength is given by

$\lambda_e = \frac{h}{p} = \frac{h}{\sqrt{2mE}}$ for electron

De-Broglie wavelength of photon is given by

$$\lambda_p = \frac{h}{p} = \frac{h}{\frac{E}{c}} = \frac{hc}{E}$$

$$\frac{\lambda_e}{\lambda_p} = \frac{1}{\sqrt{2mE}} \cdot \frac{E}{c} = \frac{1}{c}\sqrt{\frac{E}{2m}}$$

**Question No. 21**

**Single Correct Answer**
**Topic –** Mechanics
**Concept** – Laws of motion
**Subject Concept** – Circular motion
**Concept Field** – acceleration
**Question Level** - Medium
**Expected time to solve** – 45 sec

21. A disk and a sphere of same radius but different masses roll off on two inclined planes of the same altitude and length. Which one of the two objects gets to the bottom of the plane first?
    (1) Disk (2) Sphere
    (3) Both reach at the same time (4) Depends on their masses

**Sol.** **(2)**

Acceleration of the object on rough inclined plane is a = $\frac{g\sin\theta}{1+\frac{1}{mR^2}}$

For sphere $a_1 = \frac{5g\sin\theta}{7}$

For disc $a_2 = \frac{2g\sin\theta}{3}$

$a_1 > a_2$, so sphere will reach bottom first.

**Question No. 22**
**Single Correct Answer**
**Topic –** Optics
**Concept** – Ray optics
**Subject Concept** – Prism
**Concept Field** – Angle of deviation
**Question Level** - Easy
**Expected time to solve** -50 sec

22. The angle of incidence for a ray of light at a refracting surface of a prism is 45°. The angle of prism is 60°. If the ray suffers minimum deviation through the prism, the angle of minimum deviation and refractive index of the material of the prism respectively, are:
    (1) $45°; \frac{1}{\sqrt{2}}$ (2) $30°; \sqrt{2}$ (3) $45°; \sqrt{2}$ (4) $30°; \frac{1}{\sqrt{2}}$

**Sol.** **(2)**

At minimum deviation $\delta_{\min} = 2i - A$

$\delta_{\min} = 2(45) - 60$

$\delta_{\min} = 30^\circ$

Refractive index of material is

$$\mu = \frac{\sin\left(\frac{\delta_{\min} + A}{2}\right)}{\sin\left(\frac{A}{2}\right)} = \frac{\sin\left(\frac{30+60}{2}\right)}{\sin\left(30^\circ\right)}$$

$$\mu = \frac{\sin 45^\circ}{\sin 30^\circ} = \frac{\frac{1}{\sqrt{2}}}{\frac{1}{2}} = \sqrt{2}$$

**Question No. 23**

**Single Correct Answer**

**Topic –** Modern physics

**Concept** – Atomic physics

**Subject Concept** – Rutherford model

**Concept Field** – Alpha particle

**Question Level** - Medium

**Expected time to solve** – 45 sec

23. When an α-particle of mass 'm' moving with velocity 'v' bombards on a heavy nucleus of charge 'Ze', its distance of closest approach from the nucleus depends on m as:

(1) $\frac{1}{m}$ (2) $\frac{1}{\sqrt{m}}$ (3) $\frac{1}{m^2}$ (4) m

**Sol.** **(1)**

At the distance of lowest approach, total K.E. of $\alpha$-particle changes to P.E. so

$$\frac{1}{2}mv^2 = \frac{KQ \cdot q}{r} = \frac{K(ze)(2e)}{r}$$

$$r = \frac{4Kze^2}{mv^2} \Rightarrow r \propto \frac{1}{m}$$

$$r \propto \frac{1}{m}$$

**Question No. 24**
**Single Correct Answer**
**Topic –** Mechanics
**Concept** – Laws of motion
**Subject Concept** – Circular motion
**Concept Field** – Tangential acceleration
**Question Level** - Medium
**Expected time to solve** – 70 sec

24. A particle of mass 10 g moves along a circle of radius 6.4 cm with a constant tangential acceleration. What is the magnitude of this acceleration if the kinetic energy of the particle becomes equal to 8 × $10^{-4}$ J by the end of the second revolution after the beginning of the motion?

(1) 0.1 $m/s^2$ (2) 0.15 $m/s^2$ (3) 0.18 $m/s^2$ (4) $\frac{0.2\,m}{s^2}$

**Sol.** **(1)**

Tangential acceleration $a_t = r\alpha$ = constant = K

$$\alpha = \frac{K}{r}$$

At the end of second revoluation angular velocity is w then

$$w^2 - w_0^2 = 2 \propto \theta$$

$$w^2 - 0^2 = 2\left(\frac{K}{r}\right)(4\pi)$$

$$w^2 = \frac{8\pi K}{r}$$

K.E. of the particle is = K. E. = $\frac{1}{2}mv^2$

K. E. = $\frac{1}{2}mr^2w^2$

K. E. = $\frac{1}{2}m\left(r^2\right)\left(\frac{8\pi K}{r}\right)$

$$8\times10^{-4} = \frac{1}{2}\times10\times10^{-3}\times6.4\times10^{-2}\times3.14\times K$$

K $= \frac{2}{6.4\times3.14} = 0.1\frac{m}{ssec^2}$

**Question No. 25**
**Single Correct Answer**
**Topic –** Thermal physics
**Concept** – Kinetic theory of gases
**Subject Concept** – RMS speed
**Concept Field** – Dependency of RMS speed of temperature
**Question Level** - Hard
**Expected time to solve** – 60 sec

25. The molecules of a given mass of a gas have r.m.s velocity of 200 $ms^{-1}$ at 27°C and 1.0 × $10^5$ $Nm^{-2}$ pressure. When the temperature and pressure of the gas are respectively, 127°C and 0.05 × $10^5$ $Nm^{-2}$, the r.m.s. velocity of its molecules in $ms^{-1}$ is:

(1) $100\sqrt{2}$ (2) $\frac{400}{\sqrt{3}}$ (3) $\frac{100\sqrt{2}}{3}$ (4) $\frac{100}{3}$

**Sol. (2)**

Rms speed of molecules is $V_{ms} = \sqrt{\frac{3RT}{M}}$

So it depends only on temperature

$$V_{rms} \propto \sqrt{T}$$

$$\frac{V_1}{V_2} = \sqrt{\frac{T_1}{T_2}} \Rightarrow \frac{200}{V_2} = \sqrt{\frac{300}{400}}$$

$$\frac{200}{V_2} = \frac{\sqrt{3}}{2} \Rightarrow V_2 = \frac{400}{\sqrt{3}} \text{ m/sec}$$

**Question No. 26**
**Single Correct Answer**
**Topic –** Electromagnetism
**Concept** – Electromagnetic induction
**Subject Concept** – Current carrying conductor
**Concept Field** – Ampere's law
**Question Level** - Medium
**Expected time to solve** – 45 sec

26. A long straight wire of radius a carries a steady current I. the current is uniformly distributed over its cross - section. The ratio of the magnetic fields B and B′, at radial distances $\frac{a}{2}$ and 2a respectively, from the axis of the wire is:

(1) $\frac{1}{4}$ (2) $\frac{1}{2}$ (3) 1 (4) 4

**Sol.** **(3)**

$$\frac{\text{Inside the wire}}{\text{Byampere'slaw}}$$

$$\int \vec{B}\cdot d\vec{l} = \mu_0 (i_{enclosed})$$

$$\int B \cdot d\cos 0 = \mu_0 \left(\frac{I}{\pi a^2}\cdot \pi\left(\frac{a}{2}\right)^2\right)$$

$$B\int dl = \mu_0 \frac{I}{4}$$

$$B\left(2\pi\left(\frac{a}{2}\right)\right) = \frac{\mu_0 I}{4}$$

B = $\frac{\mu_0 I}{4\pi a}$

Outside the wire,

B′ = $\frac{\mu_0 I}{2\pi r} = \frac{\mu_0 I}{2\pi(2a)} = \frac{\mu_0 I}{4\pi a}$

So, $\frac{B}{B'} = 1$

**Question No. 27**

**Single Correct Answer**

**Topic –** Mechanics

**Concept** – Laws of motion

**Subject Concept** – Distance and displacement

**Concept Field** – Velocity

**Question Level** - Easy

**Expected time to solve** -30 sec

27. A particle moves so that its position vector is given by $\vec{r}$ = cos ω t $\hat{x}$ + sinωt $\hat{y}$. Where ω is a constant.

Which of the following is true?

(1) Velocity and acceleration both are perpendicular to $\vec{r}$.

(2) Velocity and acceleration both are parallel to $\vec{r}$.

(3) Velocity is perpendicular to $\vec{r}$ and acceleration is directed towards the origin.

(4) Velocity is perpendicular to $\vec{r}$ and acceleration is directed away from the origin.

**Sol.** **(3)**

Position vector is $\vec{r} = \cos\omega + x + \sin\omega + y$

Velocity of particle is $\vec{v} = \frac{d\vec{r}}{dt}$

$\vec{v} = \sin \omega t . \omega \hat{x} + \cos \omega t \cdot \omega \hat{y}$

$\vec{v} = \omega(-\sin \omega t \hat{x} + \cos \omega t \hat{y})$

Acceleration of the particle is

$\vec{a} = \frac{d\vec{v}}{dt}$

$\vec{a} = -\omega^2 (\cos \omega t x + \sin \omega t y)$

$\vec{a} = -\omega^2 \vec{r}$

So direction of $\vec{r}$ and $\vec{a}$ are opposite.

$\vec{v} \cdot \vec{a} = 0 \Rightarrow \vec{V} \perp \vec{a}$

$\vec{v} \cdot \vec{r} = 0 \Rightarrow \vec{v} \perp \vec{r}$

So, ans is (Velocity is perpendicular to $\vec{r}$ and acceleration is directed towards the origin.)

**Question No. 28**

**Single Correct Answer**

**Topic –** Mechanics

**Concept** – Laws of motion

**Subject Concept** – Circular motion

**Concept Field** – Minimum velocity

**Question Level** - Easy

**Expected time to solve** -35 sec

28. What is the minimum velocity with which a body of mass m must enter a vertical loop of radius R so that it can complete the loop?

(1) $\sqrt{gR}$ (2) $\sqrt{2gR}$ (3) $\sqrt{3gR}$ (4) $\sqrt{5gR}$

**Sol.** **(4)**

Minimum velocity required is v = $\sqrt{5gR}$

**Question No. 29**

**Single Correct Answer**
**Topic –** Modern physics
**Concept** – Photo electric effect
**Subject Concept** – Wavelength
**Concept Field** – Threshold wavelength
**Question Level** - Medium
**Expected time to solve** – 45 sec

29. When a metallic surface is illuminated with radiation of wavelength λ, the stopping potential is V. If the same surface is illuminated with radiation of wavelength 2λ, the stopping potential is $\frac{V}{4}$. The threshold wavelength for the metallic surface is:

(1) 4λ (2) 5λ (3) $\frac{5}{2}\lambda$ (4) 3λ

**Sol.** **(4)**

In photo electric effects

$eV_0 = 48 - W$

$$eV_0 = \frac{hc}{\lambda} - W$$

$$eV = \frac{hc}{\lambda} - W \quad ...(i)$$

$$e\frac{V}{4} = \frac{hc}{2\lambda} - W \quad ...(ii)$$

From (i) and (ii)

$$\frac{hc}{\lambda} - W = 4\left(\frac{hc}{2\lambda} - W\right)$$

$$\frac{hc}{\lambda} - W = \frac{2hc}{\lambda} - 4W$$

$$3W = \frac{hc}{\lambda} \Rightarrow W = \frac{hc}{3\lambda}$$

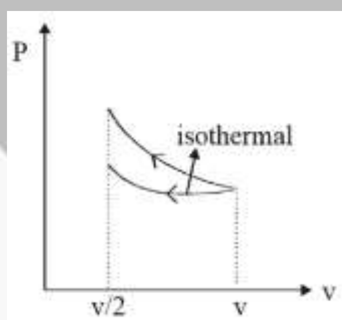
$\frac{hc}{\lambda_{max}} = \frac{hc}{3\lambda} \Rightarrow \lambda_{max}$ = threshold wavelength $3\lambda$

**Question No. 30**
**Single Correct Answer**
**Topic –** Thermal physics
**Concept** – Thermodynamics
**Subject Concept** – Thermodynamic process

**Concept Field** – Work done
**Question Level** - Medium
**Expected time to solve** – 45 sec

30. A gas is compressed isothermally to half its initial volume. The same gas is compressed separately through an adiabatic process until its volume is again reduced to half. Then:
(1) Compressing the gas isothermally will require more work to be done.
(2) Compressing the gas through adiabatic process will require more work to be done.
(3) Compressing the gas isothermally or adiabatically will require the same amount of work.
(4) Which of the case (whether compression through isothermal or through adiabatic process) requires more work will depend upon the atomicity of the gas.

**Sol.** **(2)**




Isothermal curve lie below the adiabatic curve, So in adiabatic process more work to be done.

**Question No. 31**
**Single Correct Answer**
**Topic –** Electromagnetism
**Concept** – Current electricity
**Subject Concept** – Measuring instrument
**Concept Field** – EMF
**Question Level** - Easy
**Expected time to solve** -45 sec

31. A potentiometer wire is 100 cm long and a constant potential difference is maintained across it. Two cells are connected in series first to support one another and then in opposite direction. The balance points are obtained at 50 cm and 10 cm from the

positive end of the wire in the two cases. The ratio of emf's is:

(1) 5 : 1 (2) 5 : 4 (3) 3 : 4 (4) 3 : 2

**Sol.** **(4)**


$$\frac{E_1 + E_2 = \lambda 50}{E_1 - E_2 = \lambda 10}$$

$E_1 + E_2 = 5E_1 - 5E_2$

$6E_2 = 4E_1$

$$\frac{3}{2} = \frac{E_1}{E_2}$$

**Question No. 32**

**Single Correct Answer**

**Topic –** Optics

**Concept** – Ray optics

**Subject Concept** – Telecscope

**Concept Field** – Separation between lens

**Question Level** - Easy

**Expected time to solve** -45 sec

32. A astronomical telescope has objective and eyepiece of focal lengths 40 cm and 4 cm respectively. To view an object 200 cm away from the objective, the lenses must be separated by a distance:

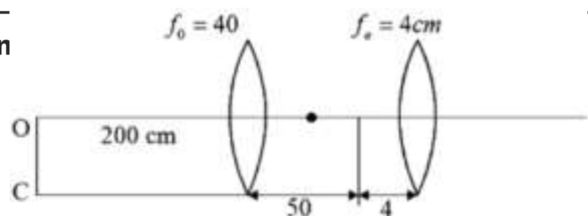
(1) 37.3 cm (2) 46.0 cm (3) 50.0 cm (4) 54.0 cm

**Sol.** **(4)**

$$\frac{1}{V} - \frac{1}{-200} = \frac{1}{40}$$

$$\frac{1}{V} = \frac{5}{5}\frac{1}{40} - \frac{1}{200}$$

$$= \frac{5}{200} - \frac{1}{200}$$



$$\frac{1}{V} = \frac{4}{200} = \frac{1}{50}$$

V = 50

$\therefore$ d = 50 + 4 = 54 cm

**Question No. 33**

**Single Correct Answer**

**Topic –** Solid and fluid mechanics

**Concept** – Fluid mechanics

**Subject Concept** – Density of liquid

**Concept Field** – Height of liquid

**Question Level** - Hard

**Expected time to solve** – 60 sec

33. Two non-mixing liquids of densities ρ and nρ (n > 1) are put in a container. The height of each liquid is h. A solid cylinder of length L and density d is put in this container. The cylinder floats with its axis vertical and length pL (p < 1) in the denser liquid. The density d is equal to:

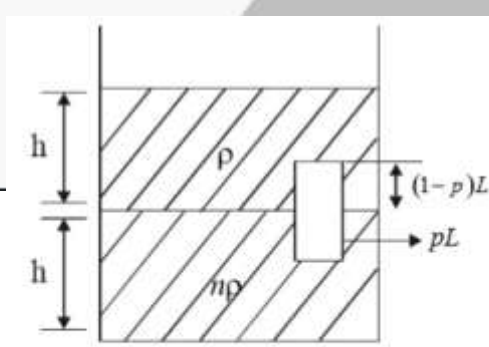
(1) {1 + (n + 1)p}ρ

(2) {2 + (n + 1)p}ρ

(3) {2 + (n − 1)p}ρ

(4) {1 + (n − 1)p}ρ

**Sol.** **(4)**

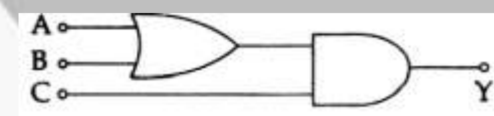



$f_0 = mg$

PA(1 − P)Lg + nρApLg = dALg

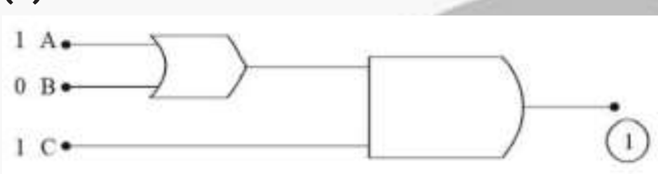
$\rho(1 - p) + n\rho p = d$
$[1 - p + np]\rho = d$
$[1 + (n - 1)p]\rho = d$

**Question No. 34**
**Single Correct Answer**
**Topic –** Modern physics
**Concept** – Logic gates
**Subject Concept** – Combination of gates
**Concept Field** – Boolean equation
**Question Level** - Easy
**Expected time to solve** - 40 sec
34. To get output 1 for the following circuit, the correct choice for the input is:




(1) A = 0, B = 1, C = 0
(2) A = 1, B = 0, C = 0
(3) A = 1, B = 1, C = 0
(4) A = 1, B = 0, C = 1

**Sol.** **(4)**




**Question No. 35**
**Single Correct Answer**
**Topic –** Thermal physics
**Concept** – Calorimetry

**Subject Concept** – Heat capacity
**Concept Field** – Latent heat of ice
**Question Level** - Medium
**Expected time to solve** – 45 sec

35. A piece of ice falls from a height h so that it melts completely. Only one-quarter of the heat produced is absorbed by the ice and all energy of ice gets converted into heat during its fall. The value of h is:
[Latent heat of ice is $3.4 \times 1^5$ J/kg and g = 10 N/kg]
(1) 34 km (2) 544 km (3) 136 km (4) 68 km

**Sol.** **(3)**

$$\frac{1}{4}\text{ mgh} = \text{mL}$$

$$h = \frac{4L}{g} = \frac{4 \times 3.4 \times 10^5}{10} = 13.6 \times 10^4$$

$= 136 \times 10^3$ km
$= 136$ km

**Question No. 36**
**Single Correct Answer**
**Topic –** Gravitation
**Concept** – Gravitation field
**Subject Concept** – Escape velocity
**Concept Field** – Escape velocity from earth
**Question Level** - Hard
**Expected time to solve** -60 sec

36. The ratio of escape velocity at earth (υe) to the escape velocity at a planet (υp) whose radius and mean density are twice as that of earth is:
(1) 1 : 2 (2) $1 : 2\sqrt{2}$ (3) 1 : 4 (4) $1 : \sqrt{2}$

**Sol.** **(2)**

$$\frac{V_e}{V_P} = \frac{\sqrt{2\frac{GM_e}{R_e}}}{\sqrt{2\frac{GM_P}{R_P}}} = \sqrt{\frac{M_R}{M_P}\frac{R_P}{R_e}} = \sqrt{\frac{P_e \frac{4}{3}\pi R_e^3 R_P}{P\frac{4}{3}\pi R_P^3 R_e}}$$

$$\frac{V_e}{V_P} = \sqrt{\frac{P_e R_e^2}{P_P R_P^2}} = \sqrt{\frac{1}{22^2}} = \frac{1}{2\sqrt{2}}$$

**Question No. 37**
**Single Correct Answer**
**Topic –** basics and laboratories
**Concept** – Basic mathematics
**Subject Concept** – Vector
**Concept Field** – Addition of vector
**Question Level** – Easy
**Expected time to solve** - 40 sec

37. If the magnitude of sum of two vectors is equal to the magnitude of difference of the two vectors, the angle between these vectors is:
(1) 0° (2) 90° (3) 45° (4) 180°

**Sol.** **(2)**

$$|\vec{A}+\vec{B}|=|\vec{A}-\vec{B}|$$

$A^2 + B^2 + 2AB \cos\theta = A^2 + B^2 = 2AB \cos\theta$

$4AB \cos\theta = 0$

$\cos\theta = 0$

$\theta = 90$

**Question No. 38**
**Single Correct Answer**
**Topic –** Modern physics
**Concept** – Atomic physics
**Subject Concept** – Balmer series
**Concept Field** – Rydberg constant
**Question Level** - Medium
**Expected time to solve** – 50 sec

38. Given the value of Rydberg constant is $10^7$ $m^{-1}$, the wave number of the last line of the Balmer series in hydrogen spectrum will be:
(1) $0.025 \times 10^4$ $m^{-1}$ (2) $0.5 \times 10^7$ $m^{-1}$ (3) $0.25 \times 10^7$ $m^{-1}$ (4) $2.5 \times 10^7$ $m^{-1}$

**Sol.** **(3)**

$$\frac{1}{\lambda} = R = \left(\frac{1}{h_1^2} - \frac{1}{h_2^2}\right)$$

Wavelength = $\frac{1}{\lambda} = R\left[\frac{1}{2^2}\right] = \frac{R}{4} = \frac{10^7}{4} = 0.25 \times 10^7\ m^{-1}$

**Question No. 39**

**Single Correct Answer**
**Topic –** Mechanics
**Concept** – Work, energy and power
**Subject Concept** – Variable force
**Concept Field** – Power
**Question Level** - Hard
**Expected time to solve** -60 sec

39. A body of mass 1 kg begins to move under the action of a time dependent force $\vec{F} = \left(2t\hat{i} + 3t^2\hat{J}\right)$ N, where $\hat{I}$ and $\hat{J}$ are unit vectors along x and y axis. What power will be developed by the force at the time t?
(1) $(2t^2 + 3t^3)$ W
(2) $(2t^2 + 4t^4)$ W
(3) $(2t^3 + 3t^4)$ W
(4) $(2t^3 + 3t^5)$ W

**Sol.** **(4)**

$$\vec{a} = 2t\hat{i} + 3t^2\hat{\o}$$

$$\vec{V} = 2t^2\hat{i} + \frac{3}{3}t^3\hat{\o}$$

$$\vec{F} = 2t\hat{i} + 3t^2\hat{\o}$$

$$P = \vec{F}\cdot\vec{V} = 2t^3 + 3t^5$$

**Question No. 40**
**Single Correct Answer**
**Topic –** Electromagneitsm
**Concept** – Current electricity
**Subject Concept** – RLC circuit
**Concept Field** – Power loss
**Question Level** - Medium
**Expected time to solve** – 45 sec

40. An inductor 20 mH, a capacitor 50 μF and a resistor 40 Ω are connected in series across a source of emf V = 10 sin 340 t. The power loss in A.C. circuit is :
(1) 0.51 W (2) 0.67 W (3) 0.76 W (4) 0.89 W

**Sol.** **(1)**

$$wL = 340 \times 20 \times 10^{-3} = 68 \times 10^{-1} = 6.8$$

$$\frac{1}{wC} = \frac{1}{340 \times 50 \times 10^{-6}} = \frac{10^4}{34 \times 5} = \frac{2}{34} \times 10^3$$

$$= 0.0588 \times 10^3 = 58.82$$

$$2=\sqrt{\left(WL-\frac{1}{wc}\right)^2+R^2}$$

$$2=\sqrt{2704+1600}\approx 65.6$$

$$i=\frac{V}{2},\frac{10}{65\times\sqrt{2}}=\frac{10}{65.6\sqrt{2}}$$

Power = $\frac{100\times 40}{(65.6)^2\times 2}=\frac{2000}{(65.6)^2}$

= 0.51 w

**Question No. 41**

**Single Correct Answer**

**Topic –** Mechanics

**Concept** – Moiton in plane

**Subject Concept** – Distance and displacement

**Concept Field** –

**Question Level** – Medium

**Expected time to solve** – 45 sec

41. If the velocity of a particle is υ = At + Bt$^2$, where A and B are constants, then the distance travelled by it between 1s and 2s is:

(1) $\frac{3}{2}A+4B$ (2) 3A + 7B (3) $\frac{3}{2}A+\frac{7}{3}B$ (4) $\frac{A}{2}+\frac{B}{3}$

**Sol. (3)**

$$V=At+Bt^2$$

$$X=\frac{At^2}{2}+\frac{Bt^3}{3}$$

t = 1

$$X_1=\frac{A}{2}+\frac{B}{3}$$

t = 2

$$X_2=2A+\frac{8B}{3}$$

$$X_2-X_1=\frac{3A}{2}+\frac{7B}{3}$$

**Question No. 42**

**Single Correct Answer**

**Topic –** Electromagnetism

**Concept** – Current electricity

**Subject Concept** – Solenoid
**Concept Field** – Self inductance of solenoid
**Question Level** – Easy
**Expected time to solve** – 40 sec

42. A long solenoid has 1000 turns. When a current of 4A flows through it the magnetic flux linked with each turn of the solenoid is $4 \times 10^{-3}$ Wb. The self-inductance of the solenoid is:

(1) 4 H (2) 3 H (3) 2 H (4) 1 H

**Sol.** **(4)**

$\phi = Li$

$1000 \times 4 \times 10_{-3} = L4$

$1 = L$

**Question No. 43**
**Single Correct Answer**
**Topic –** Electormagneitsm
**Concept** – Capacitor and capacitance
**Subject Concept** – Phasor diagram
**Concept Field** – phase difference
**Question Level** - Easy
**Expected time to solve** – 40 sec

43. A small signal voltage $V(t) = V_0 \sin \omega t$ is applied across an ideal capacitor C:

(1) Current I(t), legs voltage V(t) by 90°

(2) Over a full cycle the capacitor C does not consume any energy from the voltage source

(3) Current I(t) is in phase with voltage V(t).

(4) Current I(t) leads voltage V(t) by 180°

**Sol.** **(2)**

In capacitor current leads the voltage. Average power dissipated in capacitor is zero

**Question No. 44**
**Single Correct Answer**

**Topic –** Optics
**Concept** – Ray optics
**Subject Concept** – Mirror
**Concept Field** – Magnification
**Question Level** - Medium
**Expected time to solve** – 50 sec

44. Match the corresponding entries of column 1 with column 2. [Where m is the magnification produced by the mirror]

| | Column 1 | | Column 2 |
|---|---|---|---|
| (A) | m = -2 | (a) | Convex mirror |
| (B) | m = $-\frac{1}{2}$ | (b) | Concave mirror |
| (C) | m = +2 | (c) | Real image |
| (D) | m = $+\frac{1}{2}$ | (d) | Virtual image |

(1) A → b and c; B→ b and c; C→ b and d; D→ a and d
(2) A → a and c; B→ a and d; d→ a and b; D→ c and d
(3) A → a and d; B→ b and c; C→ b and d; D→ b and c
(4) A → c and d; B→ b and d; C→ b and c; D→ a and d

**Sol.** **(1)**

m = $\frac{-V}{u} = \frac{f}{f \times u}$

m = −2 then "V" and "u" same given

$-2 = \frac{f}{f \times u} - 2f + 2u = f$

= 3f = −2u

$\frac{+3f}{2} = 4$

For mirror so 4 negative

∴ V has to be negative

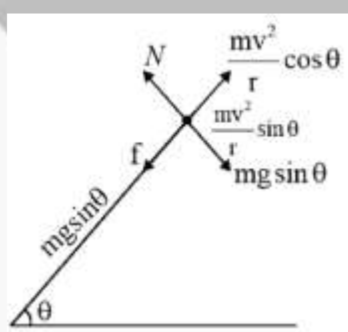
**Question No. 45**
**Single Correct Answer**
**Topic –** mechanics
**Concept** – Laws of motion and friction
**Subject Concept** – Circular motion

**Concept Field** – Minimum safe velocity
**Question Level** - Hard
**Expected time to solve** - 60 sec

45. A car is negotiating a curved road of radius R. The road is banked at an angle θ. The coefficient of friction between the tyres of the car and the road is $\mu_s$. The maximum safe velocity on this road is:

(1) $\sqrt{gR^2 \frac{\mu_S + \tan\theta}{1-\mu_s \tan\theta}}$ (2) $\sqrt{gR \frac{\mu_s + \tan\theta}{1-\mu_s \tan\theta}}$ (3) $\sqrt{\frac{g}{R} \frac{\mu_s + \tan\theta}{1-\mu_s \tan\theta}}$ (4) $\sqrt{\frac{g}{R^2} \frac{\mu_S + \tan\theta}{1-\mu_s \tan\theta}}$

**Sol.** **(2)**




$$N = mg\cos\theta + \frac{mv^2}{r}\sin\theta$$

$$f_{max} = \mu mg\cos\theta + \frac{\mu mv^2}{r}\sin\theta$$

$$mg\sin\theta + \mu mg\cos\theta + \frac{\mu mv^2}{r}\sin\theta = \frac{mv^2}{r}\cos\theta$$

$$g\sin\theta + g\cos\theta = \frac{V^2}{r}(\cos\theta - \mu\sin\theta)$$

$$gr\left[\frac{\tan\theta + \mu}{1+\mu\tan\theta}\right] = V^2$$

Question Type: NEET
Difficulty of question : Moderate
Expected time to solve : 30 sec.
Topic: Inorganic Chemistry
Concept : p-block

**136.** Hot concentrated sulphuric acid is a moderately strong oxidizing agent. Which of the following reactions does not show oxidizing behaviour ?

(1) $C + 2H_2SO_4 \rightarrow CO_2 + 2SO_2 + 2H_2O$

(2) $CaF_2 + H_2SO_4 \rightarrow CaSO_4 + 2HF$

(3) $Cu + 2H_2SO_4 \rightarrow CuSO_4 + SO_2 + 2H_2O$

(4) $3S + 2H_2SO_4 \rightarrow 3SO_2 + 2H_2O$

**Sol.** (2) $CaF_2 + H_2SO_4 \rightarrow CaSO_4 + 2HF$ In this reaction, oxidation number of none of the atom is not changed. Hence H2SO4 is not acting as oxidising agent.

Question Type: NEET
Difficulty of question : Easy
Expected time to solve: 25 sec.
Topic: Inorganic Chemistry
Concept : Coordination Compound

**137.** Which of the following pairs of d-orbitals will have electron density along the axes ?

(1) $d_{z^2}, d_{x^2-y^2}$ (2) $d_{xy}, d_{x^2-y^2}$ (3) $d_{z^2}, d_{xz}$ (4) $d_{xz}, d_{yz}$

**Sol.** (1)

$d_{z^2}$ and $d_{x^2-y^2}$ has electron density concentrated on the axis.


Question Type: NEET
Difficulty of question : Moderate
Expected time to solve : 25 sec.
Topic: Inorganic Chemistry

Concept: p-block

**138.** The correct geometry and hybridizatio0n for $XeF_4$ are :

(1) Planar triangle, $sp^3d^3$ (2) Square planar, $sp^3d^2$

(3) Octahedral, $sp^3d^2$ (4) Trigonal bipyramidal, $sp^3d$

**Sol.** (3)

$XeF_4$, $AB_4L_2 \rightarrow sp^3d^2$

$\rightarrow$ geometry $\rightarrow$ octahedral

$\rightarrow$ shape $\rightarrow$ square planar

Question Type: NEET
Difficulty of question : Moderate
Expected time to solve: 35 sec.
Topic: Inorganic Chemistry
Concept: p-block

**139.** Among the following which one is a wrong statement ?

(1) $SeF_4$ and $CH_4$ have same shape

(2) $I_3^+$ has bent geometry

(3) $PH_5$ and $BiCl_5$ do not exist

(4) $p\pi$-$d\pi$ bonds are present in $SO_2$

**Sol.** (1)

(1) $SeF_4$ –$sp^3d$, lp = 1, shape = see-saw

$CH_4$ –$sp^3$, lp = 0, shape = tetrahedral

Question Type: NEET
Difficulty of question : Easy
Expected time to solve : 25 sec.
Topic : Inorganic Chemistry
Concept : Coordination Compound

**140.** The correct increasing order of trans-effect of the following species is :

(1) $Br^- > CN^- > NH_3 > C_6H_5^-$

(2) $CN^- > Br^- > C_6H_5^- > NH_3$

(3) $NH_3 > CN^- > Br^- > C_6H_5^-$

(4) $CN^- > C_6H_5^- > Br^- > NH_3$.

**Sol.** (4)

Trans effect order – – – > > > CN C H Br NH 6 5 3

Question Type: NEET
Difficulty of question: Moderate
Expected time to solve: 30 sec.

Topic: Inorganic Chemistry
Concept: d & f-block

**141.** Which one of the following statements related to lanthanons is incorrect ?
(1) All the lanthanons are much more reactive than aluminium
(2) Ce(+4) solutions are widely used as oxidizing agent in volumetric analysis
(3) Europium shows +2 oxidation state.
(4) The basicity decreases as the ionic radius decreases from Pr to Lu.

**Sol.** (1)
(1) Lanthanon's are less reactive than aluminium due to high IP (Lanthanoid contraction) (2) $Ce^{+4}$ is good oxidising agent and easily converted into $Ce^{+3}$
(3) Eu(63) = $4f^7\ 5d^0\ 6s^2$, $Eu^{+2} = 4f^7$
(4) In lanthanoids series 'Ce' to Lu ionic radius regular decreases and covalent character increase, basic character of hydroxide decrease

Question Type: NEET
Difficulty of question: Moderate
Expected time to solve: 30 sec.
Topic: Inorganic Chemistry
Concept: Coordination Chemistry

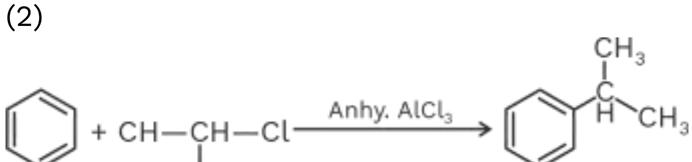
**142.** Jahn-Teller effect not observed in high spin complexes of :
(1) $d^4$ (2) $d^9$ (3) $d^7$ (4) $d^8$

**Sol.** (4)
John Teller effect explain axial distortion in unsymmetrical octahedral geometry. It is not found in high spin complexes of $d^8$.

Question Type: NEET
Difficulty of question : Moderate
Expected time to solve : 35 sec.
Topic : Organic Chemistry
Concept : Hydrocarbons

**143.** Which of the following can be used as the halide component for Friedel-Crafts reaction ?
(1) Chloroethene (2) Isopropyl chloride
(3) Chlorobenzene (4) Bromobenzene

**Sol.** (2)

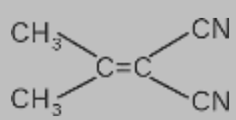


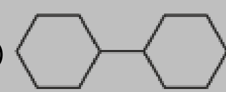
cumene

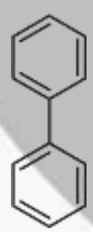
But in chlorobenzene, Bromobenzene, chloroethene lone pair of halogen are delocalised with $\pi$ bonds, so attain double bond character.

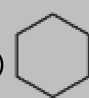
Question Type: NEET
Difficulty of question : Moderate
Expected time to solve : 30 sec.
Topic: Organic Chemistry
Concept: Stereochemistry (Isomerism)

**144.** In which of the following molecules, all atoms are coplanar ?

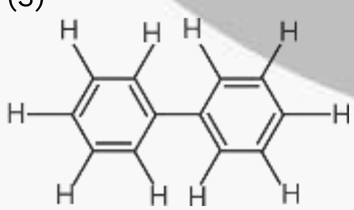
(1) 

(2) 

(3) 

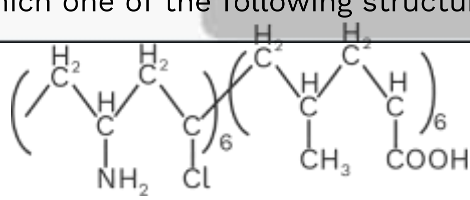
(4) 

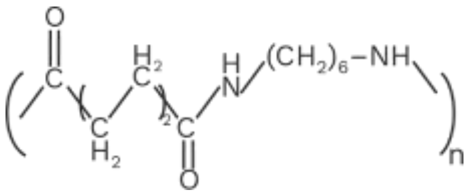
**Sol.** (3)

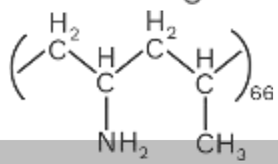


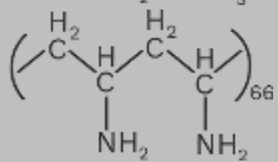
Question Type: NEET
Difficulty of question : Moderate
Expected time to solve : 35 sec.
Topic : Organic Chemistry
Concept : Polymer

**145.** Which one of the following structures represents nylon 6,6 polymer ?

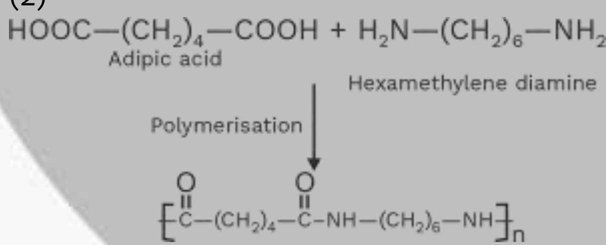
(1) 

(2) 

(3) 

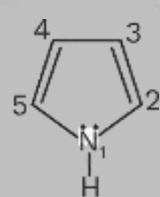
(4) 

**Sol.** (2)




Question Type: NEET
Difficulty of question : Moderate
Expected time to solve : 30 sec.
Topic : Organic Chemistry
Concept : Amines

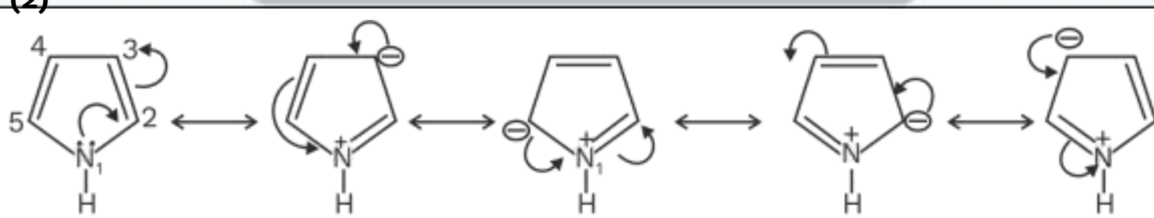
**146.** In pyrrole



The electron density is maximum on :

(1) 2 and 4 (2) 2 and 5 (3) 2 and 3 (4) 3 and 4

**Sol.** **(2)**



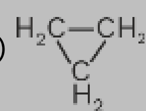
(I) (II) (III) (IV) (V)

Maximum electron density at (2) and (5)

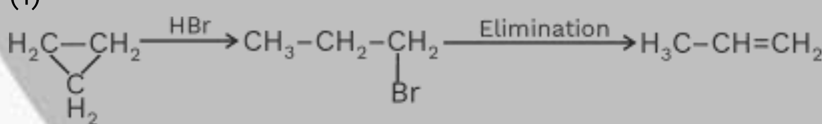
As resonating structures III and IV are more stable than (II) and (V) so are major contributor.

Question Type: NEET
Difficulty of question : Hard
Expected time to solve : 40 sec.
Topic: Organic Chemistry
Concept: Hydrocarbons

**147.** Which of the following compounds shall not produced propene by reaction with HBr followed by elimination of direct only elimination reaction ?

(1) $H_2C=C=O$ (2) $H_3C-\overset{H_2}{C}-CH_2Br$ (3)  (4) $H_2C-\overset{H_2}{C}-CH_2Br$

**Sol.** (1)



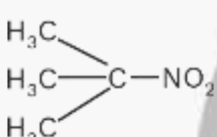
$CH_3-CH_2-CH_2-OH \xrightarrow{HBr} \xrightarrow{Elimination} H_3C-CH=CH_2$

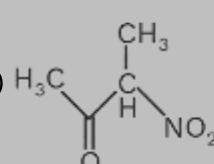
$CH_2=C=O \xrightarrow{HBr} H_2C=\underset{}{\overset{Br}{\overset{|}{C}}}-OH$ $H_3C-\overset{O}{\overset{||}{C}}-Br$

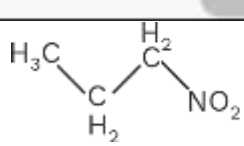
$CH_3-CH_2-CH_2-Br \xrightarrow{Elimination} CH_3-CH=CH_2$

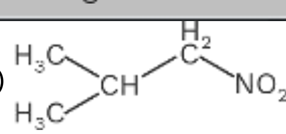
Question Type: NEET
Difficulty of question: Moderate
Expected time to solve: 30 sec.
Topic: Organic Chemistry
Concept: Amines

**148.** Which one of the following nitro-compounds does not react with nitrous acid ?

(1) 

(2) 

(3) 

(4) 

**Sol.** (1)

3°–Nitro compound does not react with $HNO_2$ because of absence of $\alpha$-H.

Question Type: NEET
Difficulty of question: Easy
Expected time to solve: 25 sec.
Topic: Organic Chemistry
Concept: Biomolecules

**149.** The central dogma of molecular genetics states that the genetic information flows from :

(1) DNA $\rightarrow$ RNA $\rightarrow$ Proteins  (2) DNA $\rightarrow$ RNA $\rightarrow$ Carbohydrates

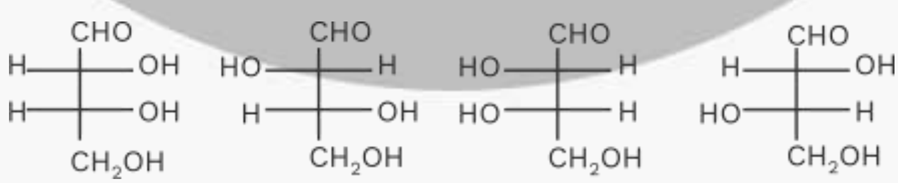
(3) Amino acids $\rightarrow$ Proteins $\rightarrow$ DNA  (4) DNA $\rightarrow$ Carbohydrates $\rightarrow$ Proteins

**Sol.** (1)

$$\text{DNA} \xrightarrow{\text{Transcription}} \text{RNA} \xrightarrow{\text{Translation}} \text{Protein}$$

Question Type: NEET
Difficulty of question: Moderate
Expected time to solve: 35 sec.
Topic: Organic Chemistry
Concept: Biomoelcules

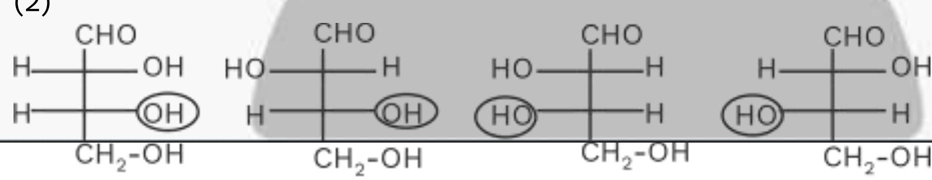
**150.** The correct corresponding order names of four aldoses with configuration given below



respectively, is :

(1) L-erythrose, L-threose, D-erythrose, D-threose
(2) D-erythrose, D-threose, L-erythrose, L-threose
(3) L-erythrose, L-threose, L-erythrose, D-threose
(4) D-threose, D-erythrose, L-threose, L-erythrose

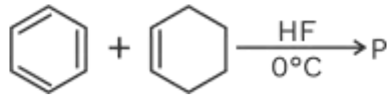
**Sol.** (2)



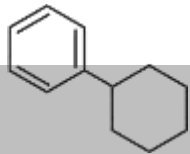
D-Erythrose  D-Therose  L-Erythrose  L-Therose

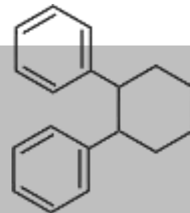
Question Type: NEET
Difficulty of question: Moderate
Expected time to solve: 35 sec.
Topic: Organic Chemistry
Concept: Hydrocarbon

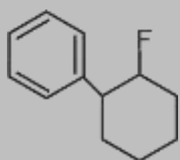
**151.** In the given reaction :

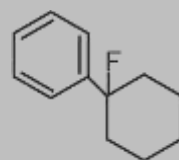



The product P is :

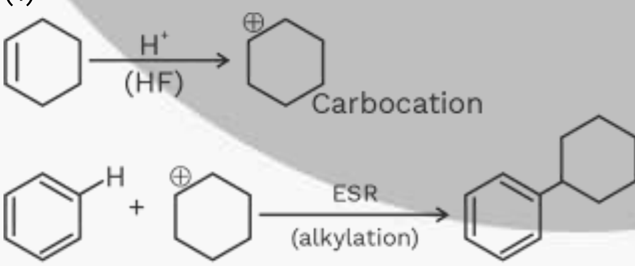
(1) 

(2) 

(3) 


(4) 

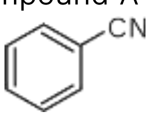

**Sol.** (1)

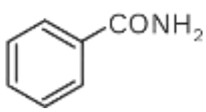


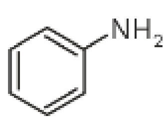

[Friedel Craft reaction]

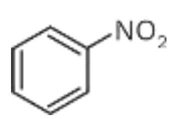
Question Type: NEET
Difficulty of question : Moderate
Expected time to solve : 30 sec.
Topic: Organic Chemistry
Concept: N-containing Compound

**152.** A given nitrogen-containing aromatic compound A reacts with Sn/HCl, followed by $HNO_2$ to give an unstable compound B. B, on treatment with phenol, forms a beautiful coloured compound C with the molecular formula $C_{12}H_{10}N_2O$. The structure of compound A is :

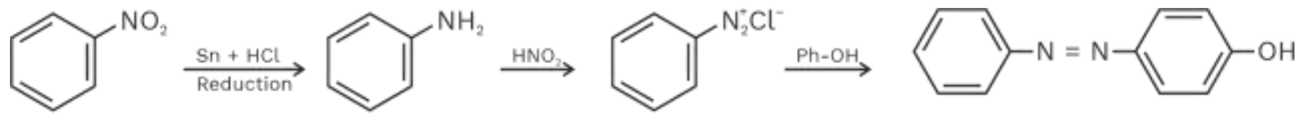
(1) 


(2) 


(3) 


(4) 


**Sol.** (4)




(A) Aniline (B) Benzene Diazonium chloride p-Hydroxy azo benzene (red colour dye)

Question Type: NEET
Difficulty of question : Moderate
Expected time to solve : 30 sec.
Topic : Organic Chemistry
Concept : Haloalkanes & Haloarenes

**153.** Consider the reaction $CH_3CH_2CH_2Br + NaCN \rightarrow CH_3CH_2CH_2CN + NaBr$

This reaction will be the fastest in :

(1) N, N'-dimethylformamide (DMF) (2) Water

(3) Ethanol (4) Methanol

**Sol.** (1)

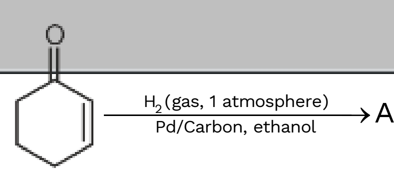
$CH_3{-}CH_2{-}CH_2Br + NaCN \rightarrow CH_3CH_2CH_2CN + NaBr$

This reaction follows by $S_N2$ path, which is favoured by polar aprotic solvents like DMF, DMSO, etc.

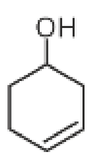
DMF (Dimethyl formamide) $H{-}\underset{\underset{O}{\|}}{C}{-}\underset{\underset{Me}{|}}{N}{-}Me$

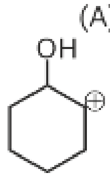
Question Type: NEET
Difficulty of question: Moderate
Expected time to solve: 30 sec.
Topic: Organic Chemistry
Concept: Aldehyde & Ketones

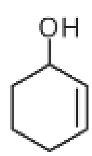
**154.** The correct structure of the product A formed in the reaction :

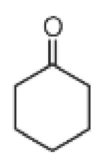



(A)

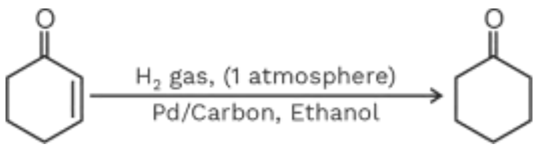
(1) 


(2) 


(3) 

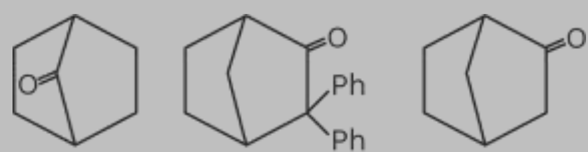

(4) 


**Sol.** (4)




Question Type: NEET
Difficulty of question : Moderate
Expected time to solve : 30 sec.
Topic: Organic Chemistry
Concept: Isomerism

**155.** Which among the given molecules can exhibit tautomerism ?




I II III

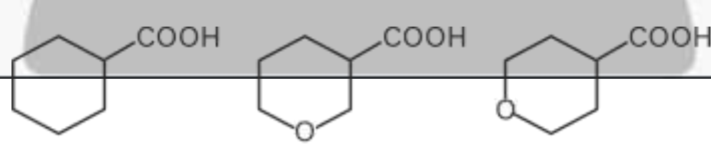
(1) Both I and II
(2) Both II and III
(3) III only
(4) Both I and III

**Sol.** (3)


Keto form Enol form

Question Type: NEET
Difficulty of question : Moderate
Expected time to solve : 35 sec.
Topic: Organic Chemistry
Concept: Carboxylic Acids

**156.** The correct order of strengths of the carboxylic acids



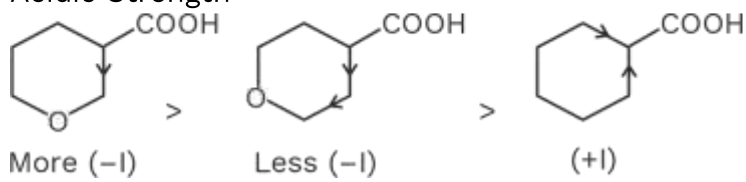

I II III

Is :

(1) III > II > I
(2) II > I > III
(3) I > II > III
(4) II > III > I

**Sol.** (4)

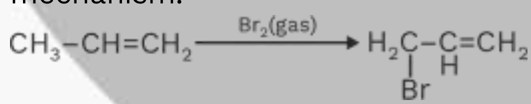
Acidic Strength




Question Type: NEET
Difficulty of question : Moderate
Expected time to solve : 30 sec
Topic : Organic Chemistry
Concept : Hydrocarbons

**157.** The compound that will react most readily with gaseous bromine has the formula :
(1) $C_4H_{10}$ (2) $C_2H_4$ (3) $C_3H_6$ (4) $C_2H_2$

**Sol.** (3)
Gaseous Bromine reacts with alkene to give allylic substituted product by free radical mechanism.




Question Type: NEET
Difficulty of question : Easy
Expected time to solve : 25 sec.
Topic : Inorganic Chemistry
Concept : Hydrogen Bonding

**158.** Which one of the following compounds shows the presence of intramolecular hydrogen bond ?
(1) Cellulose (2) Concentrated acetic acid
(3) $H_2O_2$ (4) HCN

**Sol.** (1)
In acetic acid, $H_2O_2$ and HCN inter molecular hydrogen bond present but in cellulose intramolecular hydrogen bond present.

Question Type: NEET
Difficulty of question : Moderate
Expected time to solve : 35 sec.
Topic : Physical Chemistry
Concept : Electrochemistry

**159.** The molar conductivity of a 0.5 mol/$dm^3$ solution of $AgNO_3$ with electrolytic conductivity of $5.76 \times 10^{-3}$ S $cm^{-1}$ at 298 K is :
(1) 0.086 S $cm^2$/mol (2) 28.8 S $cm^2$/mol

(3) 2.88 S $cm^2$/mol (4) 11.52 S $cm^2$/mol

**Sol.** (4)

C = 0.5 mol/$dm^3$

$\kappa = 5.76 \times 10^{-3}$ S $cm^{-1}$

T = 298 K

$$\lambda_m = \frac{\kappa \times 1000}{M} = \frac{5.76 \times 10^{-3} \times 1000}{0.5} = 11.52\ Scm^2/mol$$

Question Type: NEET
Difficulty of question : Moderate
Expected time to solve : 30 sec.
Topic : Physical Chemistry
Concept : Chemical Kinetics

**160.** The decomposition of phosphine ($PH_3$) on tungsten at low pressure is a first-order reaction. It is because the :

(1) Rate is independent of the surface coverage
(2) Rate of decomposition is very slow
(3) Rate is proportional to the surface coverage
(4) Rate is inversely proportional to the surface coverage

**Sol.** (3)

The decomposition of $PH_3$ on tungsten at low pressure is a first order reaction because rate is proportional to the surface coverage.

Question Type: NEET
Difficulty of question : Easy
Expected time to solve : 25 sec.
Topic : Physical Chemistry
Concept : Surface Chemistry

**161.** The coagulation values in millimoles per litre of the electrolytes used for the coagulation of $As_2S_3$ are given below :

I. (NaCl) = 52, II. ($BaCl_2$) = 0.69, III. ($MgSO_4$) = 0.22

The correct order of their coagulating power is :

(1) III > II > I (2) III > I > II (3) I > II > III (4) II > I > III

**Sol.** (1)

$$\text{Coagulation power} \propto \frac{1}{\text{Coagulation value}}$$

So, the order is III > II > I

Question Type: NEET
Difficulty of question : Moderate
Expected time to solve : 35 sec.
Topic : Physical Chemistry
Concept : Electrochemistry

**162.** During the electrolysis of molten sodium chloride, the time required to produce 0.10 mol of chlorine gas using a current of 3 amperes is :
(1) 220 minutes (2) 330 minutes (3) 55 minutes (4) 110 minutes

**Sol.** (4)
$2Cl^- \rightarrow Cl_2(g) + 2e^-$

$W = \frac{E}{96500} \times it$

$0.1 \times 71 = \frac{35.5}{96500} \times 3 \times t(sec)$

t (s) = 6433.33 sec.
t (min) = 107.22 min $\approx$ 110 min.

Question Type: NEET
Difficulty of question : Easy
Expected time to solve : 20 sec.
Topic : Physical Chemistry
Concept : Atomic Structure

**163.** How many electrons can fit in the orbital for which n = 3 and $l$ =1 ?
(1) 10 (2) 14 (3) 2 (4) 6

**Sol.** (3)
n = 3, l = 1 $\Rightarrow$ 3p
Total 2 electron can fit in the orbital of 3p.

Question Type: NEET
Difficulty of question : Moderate
Expected time to solve : 30 sec.
Topic : Physical Chemistry
Concept : Thermodynamics

**164.** For a sample of perfect gas when its pressure is changed isothermally from $p_i$ to $p_f$, the entropy change is given by :

(1) $\Delta S = nRT \ln\left(\frac{p_f}{p_i}\right)$ (2) $\Delta S = RT \ln\left(\frac{p_i}{p_f}\right)$

(3) $\Delta S = nR \ln\left(\frac{p_f}{p_i}\right)$ (4) $\Delta S = nR \ln\left(\frac{p_i}{p_f}\right)$

**Sol.** (4)

$$\Delta S = nC_{pm}\,\lambda n\,\frac{T_f}{T_i} + nR\ell n\frac{P_i}{P_f}$$

For isothermal $T_i = T_f$, ln1 = 0

$$\Delta S = nR\lambda n\,\frac{P_i}{P_f}$$

Question Type: NEET
Difficulty of question : Easy
Expected time to solve : 25 sec.
Topic : Physical Chemistry
Concept : Solution

**165.** The van't Hoff factor (i) for a dilute aqueous solution of the strong electrolyte barium hydroxide is :

(1) 2 (2) 3 (3) 0 (4) 1

**Sol.** (2)

$Ba(OH)_2$ is strong electrolyte, so its 100% dissociation occurs in solution

$$Ba(OH)_2 \rightarrow Ba^{+2}\,(aq) + 2OH^-\,(aq)$$

Van't Hoff factor = total number of ions present in solution i = 3

Question Type: NEET
Difficulty of question : Moderate
Expected time to solve : 30 sec.
Topic : Physical Chemistry
Concept : Ionic Equilibrium

**166.** The percentage of pyridine ($C_5H_5N$) that forms pyridinium ion ($C_5H_5N^+H$) in a 0.10 M aqueous pyridine solution ($K_b$ for $C_5H_5N = 1.7 \times 10^{-9}$) is :

(1) 0.77% (2) 1.6% (3) 0.0060% (4) 0.013%

**Sol.** (4)

Pyridine ($C_5H_5N$) is a weak base

$K_b = C\alpha^2$

$$\alpha = \sqrt{\frac{1.7\times10^{-9}}{0.1}}$$

$\alpha = 1.30 \times 10^{-4}$

$\%\alpha = 1.30 \times 10^{-4} \times 100$

$\%\alpha = 0.013\%$

Question Type: NEET
Difficulty of question : Easy
Expected time to solve : 25 sec.
Topic : Physical Chemistry
Concept : Solid State

**167.** In calcium fluoride, having the fluorite structure, the coordination numbers for calcium ion ($Ca^{2+}$) and fluoride ion ($F^-$) are :

(1) 8 and 4 (2) 4 and 8 (3) 4 and 2 (4) 6 and 6

**Sol.** (1)

In $CaF_2$, the coordination numbers for

$Ca^{+2} = 8$

$F^- = 4$

Question Type: NEET
Difficulty of question : Easy
Expected time to solve : 25 sec.
Topic : Physical Chemistry
Concept : Electrochemistry

**168.** If the $E^\circ_{cell}$ for a given reaction has a negative value, which of the following gives the correct relationships for the values of $\Delta G°$ and $K_{eq}$ ?

(1) $\Delta G° < 0; K_{eq} > 1$ (2) $\Delta G° < 0; K_{eq} < 1$

(3) $\Delta G° > 0; K_{eq} < 1$ (4) $\Delta G° > 0; K_{eq} > 1$

**Sol.** (3)

$\because \quad E^\circ_{cell} = -ve$

$\therefore \quad \Delta G° = -nF\ E^\circ_{cell}$

$\Delta G° = +ve \quad \Rightarrow \quad \Delta G > 0$

$\because \quad \Delta G° = -\ 2.303RT\log K_{eq}$

$\therefore \quad K_{eq} < 1$

Question Type: NEET
Difficulty of question : Easy
Expected time to solve : 25 sec.
Topic : Physical Chemistry
Concept : Liquid Solution

**169.** Which one of the following is incorrect for ideal solution ?

(1) $\Delta P = P_{observed} - P_{calculated\ by\ Raoult's\ law} = 0$

(2) $\Delta G_{mix} = 0$

(3) $\Delta H_{mix} = 0$

(4) $\Delta U_{mix} = 0$

**Sol.** (2)

For an ideal solution $\Delta H_{mix} = 0$

$\Delta U_{mix} = 0$

$\Delta S_{mix} \neq 0$

According to $\Delta G_{mix} = \Delta H_{mix} - T\Delta S_{mix} \Rightarrow \Delta G_{mix} \neq 0$

Incorrect answer, is $\Delta G_{mix} = 0$

Question Type: NEET
Difficulty of question : Moderate
Expected time to solve : 35 sec.
Topic : Physical Chemistry
Concept : Ionic Equilibrium

**170.** The solubility of AgCl(s) with solubility product $1.6 \times 10^{-10}$ in 0.1 M NaCl solution would be :

(1) $1.6 \times 10^{-11}$ M (2) Zero (3) $1.26 \times 10^{-5}$ M (4) $1.6 \times 10^{-9}$ M

**Sol.** (4)

| | | | | |
|---|---|---|---|---|
| NaCl(aq) | $\rightarrow$ | $Na^+(aq)$ | + | $Cl^-$ (aq) |
| 0.1M | | 0 | | 0 |
| 0 | | 0.1M | | 0.1 + S |

| | | | | |
|---|---|---|---|---|
| AgCl(s) | $\rightleftharpoons$ | $Ag^+(aq)$ | + | $Cl^-$ (aq) |
| A | | 0 | | 0 |
| a – S | | S | | S + 0.1 |

$K_{sp} = 1.6 \times 10^{-10} = [Ag^+] [Cl^-] = S (0.1 + S)$

$\because$ $K_{sp}$ is small, S is neglected with respect to 0.1 M

$1.6 \times 10^{-10} = S \times 0.1$

$S = 1.6 \times 10^{-9}$ M

Question Type: NEET
Difficulty of question : Moderate
Expected time to solve : 35 sec.
Topic : Physical Chemistry
Concept : Mole Concept

**171.** Suppose the elements X and Y combine to form two compounds $XY_2$ and $X_3Y_2$. When 0.1 mole of $XY_2$ weighs 10 g and 0.05 mole of $X_3Y_2$ weighs 9 g, the atomic weights of X and Y are :

(1) 20, 30 (2) 30, 20 (3) 40, 30 (4) 60, 40

**Sol.** (3)

Let atomic weight of x is $A_x$ and y is $A_y$.

$$n_{xyz} = 0.1 = \frac{10}{A_x + 2A_y}$$

$A_x + 2A_y = 100$ .....(1)

$$n_{x3y2} = 0.05 = \frac{9}{3A_x + 2A_y}$$

$3A_x + 2A_y = 180$......(2)

On solving eq. (1) and (2)

$A_x = 40,\ A_y = 30$

Question Type: NEET
Difficulty of question : Easy
Expected time to solve : 25 sec.
Topic : Physical Chemistry
Concept : Electrochemistry

**172.** The number of electrons delivered at the cathode during electrolysis by a current of 1 ampere in 60 seconds (charge on electron = $1.60 \times 10^{-19}$ C)

(1) $3.75 \times 10^{20}$ (2) $7.48 \times 10^{23}$ (3) $6 \times 10^{23}$ (4) $6 \times 10^{20}$

**Sol.** (1)

Q = ne

i.t = n.e

$$n = \frac{1 \times 60}{1.6 \times 10^{-19}} = 3.75 \times 10^{20} \text{ electrons}$$

Question Type: NEET
Difficulty of question : Easy
Expected time to solve : 25 sec.
Topic : Inorganic Chemistry
Concept : p-block

**173.** Boric acid is an acid because its molecule :

(1) Accepts $OH^-$ from water releasing proton
(2) Combines with proton from water molecule
(3) Contains replaceable $H^+$ ion
(4) Gives up a proton

**Sol.** (1)

$B(OH)_3 + H_2O \rightleftharpoons [B(OH)_4]^- + H^+$

Question Type: NEET
Difficulty of question : Easy
Expected time to solve : 25 sec.
Topic : Inorganic Chemistry

Concept : Coordination Compound

**174.** $AlF_3$ is soluble in HF only in presence of KF. It is due to the formation of :

(1) $AlH_3$ (2) $K[AlF_3H]$ (3) $K_3[AlF_3H_3]$ (4) $K_3[AlF_6]$

**Sol.** (4)

$AlF_3 + 3KF \rightarrow K_3[AlF_6]$

Question Type: NEET
Difficulty of question: Easy
Expected time to solve : 25 sec.
Topic : Physical Chemistry
Concept : Electrochemistry

**175.** Zinc can be coated on iron to produce galvanized iron but the reverse is not possible. It is because :

(1) Zinc has lower negative electrode potential than iron
(2) Zinc has higher negative electrode potential than iron
(3) Zinc is lighter than iron
(4) Zinc has lower melting point than iron

**Sol.** (2)

Zinc has higher negative electrode potential than iron, so iron cannot be coated on zinc.

Question Type: NEET
Difficulty of question : Easy
Expected time to solve : 25 sec.
Topic : Physical Chemistry
Concept : Surface Chemistry

**176.** The suspension of slaked lime in water is known as :

(1) Milk of lime (2) Aqueous solution of slaked lime
(3) Limewater (4) Quicklime

**Sol.** (1)

Aqueous solution of slaked lime $\Rightarrow$ Lime water
Suspension solution of slaked lime $\Rightarrow$ Milk of lime

Question Type: NEET
Difficulty of question : Moderate
Expected time to solve : 30 sec.
Topic : Inorganic Chemistry
Concept : Chemical Bonding

**177.** The hybridizations of atomic orbital of nitrogen in $NO_2^+$, $NO_3^-$ and $NH_4^+$ respectively are

(1) sp, $sp^2$ and $sp^3$ (2) $sp^2$, sp and $sp^3$

(3) sp, $sp^3$ and $sp^2$ (4) $sp^2$, $sp^3$ and sp

**Sol.** (1)

$NO_2^+ = sp$ Linear

$NO_3^- = sp^2$ Trigonal planar

$NH_4^+ = sp^3$ Tetrahedral

Question Type: NEET
Difficulty of question : Easy
Expected time to solve : 25 sec.
Topic : Inorganic Chemistry
Concept : Chemical Bonding

**178.** Which of the following fluoro-compounds is most likely to behave as a Lewis base ?

(1) $CF_4$ (2) $SiF_4$ (3) $BF_3$ (4) $PF_3$

**Sol.** (4)

$PF_3$ act as Lewis base due to present of lone pair on P-atom.

Question Type: NEET
Difficulty of question : Moderate
Expected time to solve : 30 sec.
Topic : Inorganic Chemistry
Concept : Chemical Bonding

**179.** Which of the following pairs of ions is isoelectronic and isostructural ?

(1) $SO_3^{2-}$, $CO_3^-$ (2) $ClO_3^-$, $SO_3^{2-}$ (3) $CO_3^{2-}$, $NO_3^-$ (4) $ClO_3^-$, $CO_3^{2-}$

**Sol.** (2)

(2) In $SO_3^{2-}$, $ClO_3^-$ , No. of electrons = 42,

Shape : Pyramidal

Question Type: NEET
Difficulty of question : Easy
Expected time to solve : 25 sec.
Topic : Inorganic Chemistry
Concept : s-block

**180.** In context with beryllium, which one of the following statements is incorrect ?

(1) Its salts rarely hydrolyze

(2) Its hydride is electron-deficient and polymeric

(3) It is rendered passive by nitric acid
(4) It forms $Be_2C$

**Sol.** (1)
Be salts are covalent nature, so easily hydrolyzed

## Biology 2016

Class 12th
Question type: NEET
Difficulty of question: Moderate
Expected time to solve: 40 secs
Topic: Organisms and Populations
Concept: Population Interactions
Sub-concept: Competition
Concept field: Competitive Exclusion Principle

1. Gause's principle of competitive exclusion states that
   (1) More abundant species will exclude the less abundant species through competition.
   (2) Competition for the same resources excludes species having different food preferences.
   (3) No two species can occupy the same niche indefinitely for the same limited resources.
   (4) Larger organisms exclude smaller ones through competition.

Ans. (3)
Sol. Gause's 'Competitive Exclusion Principle' states that two closely related species competing for the same resources cannot coexist indefinitely and the competitively inferior one will be eliminated eventually.

Class 12th
Question type: NEET
Difficulty of question: Moderate
Expected time to solve: 30 secs
Topic: Biotechnology and its Applications
Concept: Biotechnology Applications in Medicine
Sub-concept: GM Medicines
Concept field: Genetically Engineered Insulin

2. The two polypeptides of human insulin are linked together by
   (1) Hydrogen bonds (2) Phosphodiester bond
   (3) Covalent bond (4) Disulphide bridges

Ans. (4)
Sol. The two polypeptides called A peptide and B peptide of human insulin were produced separately using plasmids of *E.coli* and were linked together by disulphide bonds *in vitro*.

Class 12th
Question type: NEET
Difficulty of question: Easy
Expected time to solve: 30 secs

Topic: Sexual Reproduction in Flowering Plants
Concept: Post-fertilization Structure and Events
Sub-concept: Endosperm
Concept field: Free Nuclear Endosperm

3. The coconut water from tender coconut represents
   (1) Endocarp (2) Fleshy mesocarp
   (3) Free nuclear proembryo (4) Free nuclear endosperm

Ans. (4)

Sol. Endosperm development precedes embryo development. The most common type of endosperm development is free-nuclear endosperm where the PEN undergoes successive divisions without immediate cell wall formation after each division. Later on cell walls are laid between the nuclei turning it into cellular endosperm.

Class 11th
Question type: NEET
Difficulty of question: Easy
Expected time to solve: 30 secs
Topic: Biological Classification
Concept: Acellular Organisms
Sub-concept: Acellular Infectious Agents
Concept field: Viroids

4. Which of the following statements is wrong for viroids?
   (1) They lack a protein coat
   (2) They are smaller than viruses
   (3) They cause infections
   (4) Their RNA is of high molecular weight

Ans. (4)

Sol. The RNA of viroids has low molecular weight as it lacks a protein coat around it.

Class 11th
Question type: NEET
Difficulty of question: Moderate
Expected time to solve: 30 secs
Topic: Animal Kingdom
Concept: Invertebrata
Sub-concept: Arthropods
Concept field: General Characteristics

5. Which of the following features is not present in the Phylum - Arthropoda?
   (1) Chitinous exoskeleton (2) Metameric segmentation
   (3) Parapodia (4) Jointed appendages

Ans. (3)

Sol. Parapodia are lateral appendages found in aquatic annelids which help in swimming.

Class 12th
Question type: NEET
Difficulty of question: Moderate
Expected time to solve: 30 secs
Topic: Principles of Inheritance and Variation
Concept: Mendelian Disorders
Sub-concept: Sex-linked Disorders

Concept field: Haemophilia

6. Which of the following most appropriately describes haemophilia?
(1) Recessive gene disorder
(2) X - linked recessive gene disorder
(3) Chromosomal disorder
(4) Dominant gene disorder

Ans. (2)

Sol. Haemophilia is a sex linked recessive disease. The defective gene is present on the X chromosome of the 23rd pair but is recessive in nature. The heterozygous female (carrier) for haemophilia may transmit the disease to sons. The haemophilic male transmits the defective gene to the daughters.

Class 11th
Question type: NEET
Difficulty of question: Moderate
Expected time to solve: 30 secs
Topic: Photosynthesis
Concept: Photosystems
Sub-concept: Existence of PS I and PS II
Concept field: Emerson's Enhancement Effect and Red drop

7. Emerson's enhancement effect and Red drop has been instrumental in the discovery of
(1) Photophosphorylation and non-cyclic electron transport
(2) Two photosystems operating simultaneously
(3) Photophosphorylation and cyclic electron transport
(4) Oxidative phosphorylation

Ans. (2)

Sol. Emerson’s enhancement effect is the increase in the rate of photosynthesis after the chloroplast were exposed to light of wavelength 680 nm and more than 680 nm (far red spectrum) simultaneously. This proved the existence of two photosystems in plants.
He also proved that when the chloroplasts were exposed to only the monochromatic light of more than 680 nm (far red spectrum), there was a decrease in the photosynthetic rate which is known as red drop.

Class 11th
Question type: NEET
Difficulty of question: Moderate
Expected time to solve: 30 secs
Topic: Mineral Nutrition
Concept: Micro and Macronutrients
Sub-concept: Macronutrients
Concept field: Examples

8. In which of the following, all three are macronutrients?
(1) Boron, zinc, manganese
(2) Iron, copper, molybdenum
(3) Molybdenum, magnesium, manganese
(4) Nitrogen, nickel, phosphorus

Ans. (none)

Sol. In none of the groups all the elements are macronutrients.

Class 11th

Question type: NEET
Difficulty level: Easy
Expected time to solve: 20 secs
Topic: Breathing and Exchange of Gases
Concept: Respiratory System
Sub-concept: Disorders of Respiratory System
Concept field: Emphysema

9. Name the chronic respiratory disorder caused mainly by cigarette smoking
(1) Emphysema (2) Asthma
(3) Respiratory acidosis (4) Respiratory alkalosis

Ans. (1)

Sol. Emphysema is caused by excessive cigarette smoking. In this disorder the alveolar walls get damaged and become stiff, so the surface area for exchanges of gases reduces tremendously.

Class 12th
Question type: NEET
Difficulty level: Easy
Expected time to solve: 20 secs
Topic: Environmental Issues
Concept: Deforestation
Sub-concept: Agriculture
Concept field: Traditional Techniques

10. A system of rotating crops with legume or grass pasture to improve soil structure and fertility is called
(1) Ley farming (2) Contour farming
(3) Strip farming (4) Shifting agriculture

Ans. (1)

Sol. Ley farming is used to improve soil structure in a natural way.

Class 11th
Question type: NEET
Difficulty level: Easy
Expected time to solve: 20 secs
Topic: Cell-The Unit of Life
Concept: Eukaryotic Cell
Sub-concept: Cell Organelles
Concept field: Semi-autonomous Organelles

11. Mitochondria and chloroplast are
(a) semi-autonomous organelles
(b) formed by division of pre-existing organelles and they contain DNA but lack protein synthesizing machinery.

Which one of the following options is correct?
(1) Both (a) and (b) are correct (2) (b) is true but (a) is false
(3) (a) is true but (b) is false (4) Both (a) and (b) are false

Ans. (3)

Sol. 70 S ribosomes and RNA are present in both mitochondria and chloroplasts which are required for protein synthesis.

Class 12th

Question type: NEET
Difficulty level: Easy
Expected time to solve: 20 secs
Topic: Reproductive Health
Concept: Strategies
Sub-concept: Amniocentesis
Concept field: Significance

12. In the context of Amniocentesis, which of the following statements is incorrect?
(1) It is usually done when a woman is between 14-16 weeks pregnant.
(2) It is used for prenatal sex determination
(3) It can be used for detection of Down syndrome
(4) It can be used for detection of Cleft palate

Ans. (4)

Sol. Detection of Cleft palate can be done by ultrasound as early as 16 weeks into a pregnancy.

Class 11th
Question type: NEET
Difficulty level: Tough
Expected time to solve: 30 secs
Topic: Photosynthesis in Higher Plants
Concept: The Electron Transport
Sub-concept: Chemiosmotic Hypothesis
Concept field: Proton Gradient

13. In a chloroplast the highest number of protons are found in
(1) Stroma
(2) Lumen of thylakoids
(3) Inter membrane space
(4) Antennae complex

Ans. (2)

Sol. Splitting of water molecules takes place on the inner side of the membrane so the protons of hydrogen ions produced by the splitting accumulate within the lumen of thylakoids.

Class 11th
Question type: NEET
Difficulty level: Moderate
Expected time to solve: 30 secs
Topic: Neural Control and Coordination
Concept: Sense Organs
Sub-concept: Human Eye
Concept field: Anatomy of Human Eye

14. Photosensitive compound in human eye is made up of
(1) Guanosine and Retinol
(2) Opsin and Retinal
(3) Opsin and Retinol
(4) Transducin and Retinene

Ans. (2)

Sol. Photosensitive compound in the human eye is made up of a purplish-red protein called rhodopsin which contains a derivative of Vitamin A. These are located on the outermost layer of the retina.

Class 11th
Question type: NEET

Difficulty level: Easy
Expected time to solve: 20 secs
Topic: Cell Cycle and Cell Division
Concept: Mitosis Cell Division
Sub-concept: Metaphase
Concept field: Events in Metaphase

15. Spindle fibres attach on to
(1) Telomere of the chromosome
(2) Kinetochore of the chromosome
(3) Centromere of the chromosome
(4) Kinetosome of the chromosome

Ans. (2)

Sol. Kinetochores are small disc-like structures present on the surface of the centromere and these are the sites for the attachment of spindle fibres so that the chromatids can be pulled to opposite poles during anaphase stage.

Class 11th
Question type: NEET
Difficulty level: Easy
Expected time to solve: 20 secs
Topic: Animal Kingdom
Concept: Phylum Chordata
Sub-concept: Class Mammalia
Concept field: Habitat of Mammals

16. Which is the National Aquatic Animal of India?
(1) Gangetic shark (2) River dolphin (3) Blue whale (4) Seahorse

Ans. (2)

Sol. To save the dolphins from extinction the government of India has declared the river dolphin, the national aquatic animal of India.

Class 12th
Question type: NEET
Difficulty level: Easy
Expected time to solve: 20 secs
Topic: Molecular Basis of Inheritance
Concept: Regulation of Gene Expression
Sub-concept: The *Lac* Operon
Concept field: Inducible Operon

17. Which of the following is required as inducer(s) for the expression of *Lac* operon?
(1) Glucose (2) Galactose
(3) Lactose (4) Lactose and galactose

Ans. (3)

Sol. The *Lac* operon in *E. coli.* bacteria (prokaryote) remain off but if lactose is added in its surrounding medium the operon turns on and starts metabolising lactose. *Lac* operon is inducible.

Class 11th
Question type: NEET
Difficulty level: Moderate
Expected time to solve: 30 secs

Topic: Chemical Control and Coordination
Concept: Endocrine Glands
Sub-concept: Hormones
Concept field: Antagonist Hormones

18. Which of the following pairs of hormones are not antagonistic (having opposite effects) to each other?
(1) Parathormone – Calcitonin
(2) Insulin – Glucagon
(3) Aldosterone – Atrial Natriuretic Factor
(4) Relaxin – Inhibin

Ans. (4)

Sol. Relaxin is a hormone which is secreted by the placenta, it relaxes the ligaments of the pelvis to ease partirution. While inhibin is secreted from the anterior pituitary, it decreases the secretion of FSH.

Class 11th
Question type: NEET
Difficulty level: Easy
Expected time to solve: 20 secs
Topic: Cell-The Unit of Life
Concept: Cytoskeleton
Sub-concept: Microtubules
Concept field: Cell Structures Composed of Microtubules

19. Microtubules are the constituents of
(1) Cilia, Flagella and Peroxisomes
(2) Spindle fibres, Centrioles and Cilia
(3) Centrioles, Spindle fibres and Chromatin
(4) Centrosome, Nucleosome and Centrioles

Ans. (2)

Sol. Microtubules are made up of tubulin proteins. Spindle fibres, centrioles and cilia have a considerable percentage of microtubules within them.

Class 11th
Question type: NEET
Difficulty level: Easy
Expected time to solve: 20 secs
Topic: Cell-The Unit of Life
Concept: Prokaryotic Cell
Sub-concept: Cell Organelles
Concept field: Polysome

20. A complex of ribosomes attached to a single strand of RNA is known as
(1) Polysome (2) Polymer (3) Polypeptide (4) Okazaki fragment

Ans. (1)

Sol. Polysomes are formed when two or more ribosomes are attached to a complex of mRNA molecules.

Class 12th
Question type: NEET
Difficulty level: Easy
Expected time to solve: 20 secs

Topic: Human Reproduction
Concept: Female Reproductive System
Sub-concept: Fertilization
Concept field: Site of Fertilization

21. Fertilization in humans is practically feasible only if
    (1) the sperms are transported into vagina just after the release of ovum in fallopian tube
    (2) the ovum and sperms are transported simultaneously to ampullary isthmic junction of the fallopian tube
    (3) the ovum and sperms are transported simultaneously to ampullary - isthmic junction of the cervix
    (4) the sperms are transported into cervix within 48 hrs of release of ovum in uterus

Ans. (2)

Sol. The process of fusion of a sperm with an ovum is called fertilization and it occurs at the ampullary - isthmic junction of the fallopian tube if sperm and ovum are transported simultaneously to this region.

Class 11th
Question type: NEET
Difficulty level: Easy
Expected time to solve: 20 secs
Topic: Breathing and Exchange of Gases
Concept: Human Respiratory System
Sub-concept: Respiratory Disorders
Concept field: Asthma

22. Asthma may be attributed to
    (1) bacterial infection of the lungs
    (2) allergic reaction of the mast cells in the lungs
    (3) inflammation of the trachea
    (4) accumulation of fluid in the lungs

Ans. (2)

Sol. In asthma inflammation of bronchi and bronchioles occurs. It is a chronic problem so allergic reactions in the respiratory tract are more frequent which may lead to allergic reactions of the mast cells in the lungs.

Class 11th
Question type: NEET
Difficulty level: Easy
Expected time to solve: 20 secs
Topic: Plant Growth and Development
Concept: Plant Growth Regulators
Sub-concept: Discovery
Concept field: Bioassay Method

23. The Avena curvature is used for bioassay of
    (1) ABA (2) $GA_3$ (3) IAA (4) Ethylene

Ans. (3)

Sol. Bioassay is a quantitative method used to correlate the curvature obtained in a plant after the application of an external stimulus like light and the amount of growth substance produced endogenously.

Class 11th
Question type: NEET
Difficulty level: Easy
Expected time to solve: 20 secs
Topic: Morphology of Flowering Plants
Concept: Floral Leaves
Sub-concept: Carolla
Concept field: Types of Corolla

24. The standard petal of a papilionaceous corolla is also called
(1) Carina (2) Pappus (3) Vexillum (4) Corona

Ans. (3)

Sol. The papilionaceous corolla are uneven in size and one of them is largest called vexillum.

Class 11th
Question type: NEET
Difficulty level: Easy
Expected time to solve: 20 secs
Topic: Morphology of Flowering Plants
Concept: Plant Families
Sub-concept: Family Liliaceae
Concept field: Characteristic Features

25. Tricarpellary, syncarpous, gynoecium is found in flowers of
(1) Liliaceae (2) Solanaceae (3) Fabaceae (4) Poaceae

Ans. (1)

Sol. Family Liliaceae comprises monocot plants and monocots have trimerous flowers i.e., floral parts are multiple of three so is the carpel.

Class 11th
Question type: NEET
Difficulty level: Easy
Expected time to solve: 20 secs
Topic: Biological Classification
Concept: Five Kingdom Classification
Sub-concept: Fungi
Concept field: Characteristic Features

26. One of the major components of cell wall of most fungi is
(1) Chitin (2) Peptidoglycan (3) Cellulose (4) Hemicellulose

Ans. (1)

Sol. Fungal cell wall is made up of chitin, a complex polysaccharide which contains nitrogen. It is made of a linear polymer of N-acetyl-D-glucosamine monomers.

Class 12th
Question type: NEET
Difficulty level: Easy
Expected time to solve: 20 secs
Topic: Human Reproduction
Concept: Menstrual Cycle
Sub-concept: Events during Menstrual Cycle
Concept field: Follicular Phase

27. Select the incorrect statement
(1) FSH stimulates the sertoli cells which help in spermiogenesis
(2) LH triggers ovulation in ovary
(3) LH and FSH decrease gradually during the follicular phase
(4) LH triggers secretion of androgens from the Leydig cells

Ans. (3)

Sol. The reproductive cycle in the female primates is called menstrual cycle and it is repeated at the interval of 28 or 29 days. menstrual cycle is:
During Proliferative phase also called follicular phase of menstrual cycle the endometrial lining start regenerating and, in the ovary, primary follicle grows and mature into Graafian follicle simultaneously, due to rise in gonadotropins (LH and FSH).

Class 11th
Question type: NEET
Difficulty level: Easy
Expected time to solve: 20 secs
Topic: Cell Cycle and Cell Division
Concept: Meiosis Cell Division
Sub-concept: Meiosis I
Concept field: Prophase I

28. In meiosis crossing over is initiated at
(1) Pachytene (2) Leptotene (3) Zygotene (4) Diplotene

Ans. (1)

Sol. Crossing over is a major event in meiosis cell division that results in variations. It occurs in the pachytene sub phase of meiosis-I.

Class 12th
Question type: NEET
Difficulty level: Easy
Expected time to solve: 20 secs
Topic: Principles of Inheritance
Concept: Mendel’s Law of Inheritance
Sub-concept: Law of Segregation
Concept field: Purity of Gametes

29. A tall true breeding garden pea plant is crossed with a dwarf true breeding garden pea plant. When the F1 plants were selfed the resulting genotypes were in the ratio of
(1) 1 : 2 : 1 :: Tall homozygous : Tall heterozygous: Dwarf
(2) 1 : 2 : 1 :: Tall heterozygous : Tall homozygous : Dwarf
(3) 3 : 1 :: Tall : Dwarf
(4) 3 : 1 :: Dwarf : Tall

Ans. (1)

Sol. When selfing between F1 plants is done, gametes are formed. It was observed that a gamete carries either a recessive allele or a dominant allele but not both the alleles at the same time. Due to this reason, the law of segregation is known as the law of purity of gametes.

Class 12th
Question type: NEET
Difficulty level: Easy

Expected time to solve: 20 secs
Topic: Biodiversity and Conservation
Concept: Loss of Biodiversity
Sub-concept: Causes of Biodiversity Losses
Concept field: Habitat Loss and Fragmentation

30. Which of the following is the most important cause of animals and plants being driven to extinction?
(1) Over – exploitation (2) Alien species invasion
(3) Habitat loss and fragmentation (4) Co-extinctions

Ans. (3)

Sol. The conversion of forests into agricultural land and building of dams or roads in a natural habitat results in its fragmentation; this has caused extinction of animals and plants.

Class 12th
Question type: NEET
Difficulty level: Easy
Expected time to solve: 20 secs
Topic: Ecosystem
Concept: Characteristics of an Ecosystem
Sub-concept: Man-made Ecosystems
Concept field: Cropland Ecosystem

31. Which of the following is a characteristic feature of the cropland ecosystem?
(1) Absence of soil organisms (2) Least genetic diversity
(3) Absence of weeds (4) Ecological succession

Ans. (2)

Sol. The Cropland ecosystem is a man-made ecosystem, and the cultivated crops belong to one or two plant species at a time. Hence, genetic diversity is least in such ecosystems.

Class 12th
Question type: NEET
Difficulty level: Easy
Expected time to solve: 20 secs
Topic: Human Reproduction
Concept: Female Reproductive System
Sub-concept: Menstrual Cycle
Concept field: GnRH Secretion

32. Changes in GnRH pulse frequency in females is controlled by circulating levels of
(1) estrogen and progesterone (2) estrogen and inhibin
(3) progesterone only (4) progesterone and inhibin

Ans. (1)

Sol. The reproductive system is controlled by GnRH secretion from the brain which is under the influence of the amount of estrogen and progesterone is circulated. GnRH pulse frequency increases from one pulse every 90-100 min in the beginning of the follicular phase which increases in the late follicular phase.

Class 12th
Question type: NEET
Difficulty level: Easy

Expected time to solve: 20 secs
Topic: Biotechnology-Principles and Processes
Concept: Tools of Recombinant DNA Technology
Sub-concept: Cloning Vectors
Concept field: Features of Cloning Vectors

33. Which of the following is not a feature of the plasmids?
(1) Independent replication
(2) Circular structure
(3) Transferable
(4) Single - stranded

Ans. (4)

Sol. Plasmids are found in bacterial cells independent of the control of chromosomal DNA. It is a single circular strand of DNA.

Class 11th
Question type: NEET
Difficulty level: Easy
Expected time to solve: 20 secs
Topic: Structural Organisation in Animals
Concept: Cockroach
Sub-concept: Reproduction in Cockroach
Concept field: Development of Offsprings

34. Which of the following features is not present in *Periplaneta americana*?
(1) Schizocoelom as body cavity
(2) Indeterminate and radial cleavage during embryonic development
(3) Exoskeleton composed of N-acetylglucosamine
(4) Metamerically segmented body

Ans. (2)

Sol. Indeterminate and radial cleavage during embryonic development is a feature of vertebrates and certain invertebrates like echinoderms.

Class 12th
Question type: NEET
Difficulty level: Easy
Expected time to solve: 20 secs
Topic: Human Health and Diseases
Concept: Immunity
Sub-concept: Auto Immunity
Concept field: Auto-immune Diseases

35. In higher vertebrates, the immune system can distinguish self-cells and non-self. If this property is lost due to genetic abnormality and it attacks self-cells, then it leads to
(1) Allergic response
(2) Graft rejection
(3) Auto-immune disease
(4) Active immunity

Ans. (3)

Sol. In auto-immune disease the immune system mistakes the body cells as foreign and starts attacking them.

Class 12th
Question type: NEET
Difficulty level: Moderate
Expected time to solve: 40 secs

Topic: Principles of Inheritance
Concept: Mendelian Laws
Sub-concept: Exceptions to Mendelian Laws of Inheritance
Concept field: Non- Mendelian Inheritance

36. Match the terms in Column-I with their description in Column-II and choose the correct option.

| Column-I | Column-II |
|---|---|
| (a) Dominance | (i) Many genes govern a single character |
| (b) Codominance | (ii) In a heterozygous organism only one allele expresses itself |
| (c) Pleiotropy | (iii) In a heterozygous organism both alleles express themselves fully |
| (d) Polygenic inheritance | (iv) A single gene influences many characters |

| | (a) | (b) | (c) | (d) |
|---|---|---|---|---|
| (1) | (ii) | (ii) | (i) | (iii) |
| (2) | (ii) | (ii) | (iv) | (iii) |
| (3) | (iv) | (iii) | (ii) | (iv) |
| (4) | (iv) | (iii) | (i) | (ii) |

Ans. (None)

Sol. Dominance- In a heterozygous organism only one allele expresses itself. Co-dominance - In a heterozygous organism both alleles express themselves fully. Pleiotropy - A single gene influences many characters.
Polygenic inheritance - Many genes govern a single character.

Class 12th
Question type: NEET
Difficulty level: Easy
Expected time to solve: 20 secs
Topic: Environmental Issues
Concept: Conservation of Forests
Sub-concept: People's Participation in Conservation of Forests
Concept field: JFM

37. Joint Forest Management Concept was introduced in India during
(1) 1960 s (2) 1970 s (3) 1980 s (4) 1990 s

Ans. (3)

Sol. When the government of India realised the participation of local communities in conserving forests then in the 1980s the Joint Forest Management Concept was introduced.

Class 12th
Question type: NEET
Difficulty level: Moderate
Expected time to solve: 40 secs
Topic: Principles of Inheritance
Concept: Genetic Disorders
Sub-concept: Mendalian and Chromosomal Disorders
Concept field: Inheritance of Genetic Disorders

38. Pick out the correct statements.
(a) Haemophilia is a sex-linked recessive disease
(b) Down's syndrome is due to aneuploidy

(c) Phenylketonuria is an autosomal recessive gene disorder.
(d) Sickle cell anaemia is a X-linked recessive gene disorder
(1) (a) and (d) are correct (2) (b) and (d) are correct
(3) (a), (c) and (d) are correct (4) (a), (b) and (c) are correct

Ans. (4)

Sol. Sickle cell anaemia is an autosome linked recessive trait.

Class 11th
Question type: NEET
Difficulty level: Easy
Expected time to solve: 30 secs
Topic: Biological Classification
Concept: Five Kingdom Classification
Sub-concept: Bacteria
Concept field: Eubacteria

39. Which one of the following statements is wrong?
(1) Cyanobacteria are also called blue-green algae
(2) Golden algae are also called desmids
(3) Eubacteria are also called false bacteria.
(4) Phycomycetes are also called algal fungi

Ans. (3)

Sol. In latin '*eu*' means 'true'. Therefore, eubacteria means true bacteria.

Class 12th
Question type: NEET
Difficulty level: Easy
Expected time to solve: 20 secs
Topic: Reproduction in Flowering Plants
Concept: Flower
Sub-concept: Stamen
Concept field: Structure

40. Proximal end of the filament of stamen is attached to the
(1) Anther (2) Connective (3) Placenta (4) Thalamus or petal

Ans. (4)

Sol. Proximal end means the basal part of the filament and it remains attached to the thalamus.

Class 12th
Question type: NEET
Difficulty level: Moderate
Expected time to solve: 40 secs
Topic: Reproductive Health
Concept: Birth Control Methods
Sub-concept: Contraceptives
Concept field: Surgical Methods

41. Which of the following approaches does not give the defined action of contraceptive?
(1) Barrier methods - Prevent fertilization
(2) Intrauterine devices - Increase phagocytosis of sperms, suppress sperm motility and fertilizing capacity of sperms

(3) Hormonal contraceptives - Prevent/retard entry of sperms, prevent ovulation and fertilization
(4) Vasectomy - Prevents spermatogenesis

Ans. (4)

Sol. Vasectomy is cutting and ligating both the vas deferens in males to block the gamete transport.

Class 12th
Question type: NEET
Difficulty level: Easy
Expected time to solve: 20 secs
Topic: Biotechnology – Principles and Processes
Concept: Amplification of Gene of Interest
Sub-concept: PCR
Concept field: Thermostable DNA Polymerase

42. The taq polymerase enzyme is obtained from
(1) *Thermus aquaticus*
(2) *Thiobacillus ferrooxidans*
(3) *Bacillus subtilis*
(4) *Pseudomonas putida*

Ans. (1)

Sol. Thermostable DNA polymerase enzyme is obtained from a bacterium *Thermus aquaticus*.
This enzyme remains active at high temperature and is of great significance in PCR.

Class 12th
Question type: NEET
Difficulty level: Moderate
Expected time to solve: 40 secs
Topic: Human Reproduction
Concept: Menstrual Cycle
Sub-concept: Events in Menstrual Cycle
Concept field: Role of Inhibin

43. Identify the correct statement on 'inhibin'.
(1) Inhibits the secretion of LH, FSH and Prolactin.
(2) Is produced by granulosa cells in ovary and inhibits the secretion of FSH.
(3) Is produced by granulosa cells in ovary and inhibits the secretion of LH.
(4) Is produced by nurse cells in testes and inhibits the secretion of LH.

Ans. (2)

Sol. Inhibin hormone in women inhibits the secretion of FSH and reduces the hypothalamic LH.

Class 12th
Question type: NEET
Difficulty level: Easy
Expected time to solve: 20 secs
Topic: Biotechnology and its Applications
Concept: GMO
Sub-concept: Pest Resistance Plants
Concept field: GMO Plants

44. Which part of the tobacco plant is infected by *Meloidogyne incognita*?
(1) Flower (2) Leaf (3) Stem (4) Root

Ans. (4)
Sol. *Meloidogyne incognita,* is a nematode which infects the roots of tobacco plants.

Class 12th
Question type: NEET
Difficulty level: Easy
Expected time to solve: 20 secs
Topic: Human Health and Diseases
Concept: Immunity
Sub-concept: Active and Passive Immunity
Concept field: Active Immunity

45. Anti-venom injection contains preformed antibodies while polio drops that are administered into the body contain
(1) Activated pathogens
(2) Harvested antibodies
(3) Gamma globulin
(4) Attenuated pathogens

Ans. (4)
Sol. In the form of polio drops, our body is exposed to weak pathogens and to fight against it antibodies are produced. This type of immunity is called active immunity.

Class 11th
Question type: NEET
Difficulty of question: Easy
Expected time to solve: 20 secs
Topic: Cell - The Unit of Life
Concept: Eukaryotic Cell
Sub-concept: Cell Organelles
Concept field: Single Membrane bound Cell Organelles

46. Which one of the following cell organelles is enclosed by a single membrane?
(1) Mitochondria
(2) Chloroplasts
(3) Lysosomes
(4) Nuclei

Ans. (3)
Sol. Lysosomes are enclosed by a single membrane.

Class 11th
Question type: NEET
Difficulty of question: Moderate
Expected time to solve: 20 secs
Topic: Locomotion and Movement
Concept: Muscle Contraction
Sub-concept: Mechanism of Muscle Contraction
Concept field: Disorders of Muscular and Skeletal System

47. Lack of relaxation between successive stimuli in sustained muscle contraction is known as
(1) Spasm
(2) Fatigue
(3) Tetanus
(4) Tonus

Ans. (3)
Sol. Tetanus is a bacterial disease characterized by muscle spasm. There is lack of relaxation between successive stimuli in sustained muscle contraction.

Class 11th
Question type: NEET
Difficulty of question: Moderate

Expected time to solve:15 secs
Topic: Morphology of Flowering Plants
Concept: The Stem
Sub-concept: Accessory Functions of Stem
Concept field: Modification of Stem

48. Which of the following is not a stem modification?
(1) Pitcher of *Nepenthes* (2) Thorns of citrus
(3) Tendrils of cucumber (4) Flattened structures of *Opuntia*

Ans. (1)

Sol. Pitcher of *Nepenthes*, is a modification of a leaf.

Class 11th
Question type: NEET
Difficulty of question: Moderate
Expected time to solve: 20 secs
Topic: Cell - The Unit of Life
Concept: Eukaryotic Cells
Sub-concept: Cell Organelles
Concept field: Vacuole

49. Water soluble pigments found in plant cell vacuoles are
(1) Xanthophylls (2) Chlorophylls (3) Carotenoids (4) Anthocyanins

Answer (4)

Sol. A water-soluble pigment, present in sap vacuole of plants is anthocyanin.

Class 11th
Question type: NEET
Difficulty of question: Moderate
Expected time to solve: 15 secs
Topic: Plant Kingdom
Concept: Divisions in Plant Kingdom
Sub-concept: Gymnosperms
Concept field: Characteristics and Examples

50. Select the correct statement.
(1) Gymnosperms are both homosporous and heterosporous
(2) *Salvinia, Ginkgo* and *Pinus* all are gymnosperms
(3) *Sequoia* is one of the tallest trees
(4) The leaves of gymnosperms are not well adapted to extremes of climate

Ans. (3)

Sol. *Sequoia* is one of the tallest gymnosperms.

Class 12th
Question type: NEET
Difficulty of question: Difficult
Expected time to solve: 20 secs
Topic: Molecular Basis of Inheritance
Concept: DNA Fingerprinting
Sub-concept: Techniques used for DNA Fingerprinting
Concept field: Techniques used for DNA Fingerprinting

51. Which of the following is not required for any of the techniques of DNA fingerprinting available at present?

(1) Polymerase chain reaction (2) Zinc finger analysis
(3) Restriction enzymes (4) DNA–DNA hybridization

Ans. (2)

Sol. Zinc finger is a small protein structural motif that is characterized by the coordination of one or more zinc ions (Zn) in order to stabilize the fold. It is not used in DNA fingerprinting techniques.

Class 11th
Question type: NEET
Difficulty of question: Moderate
Expected time to solve: 30 secs
Topic: Structural Organisation in Animals
Concept: Animal Tissues
Sub-concept: Muscle Tissues
Concept field: Smooth Muscles

52. Which type of tissue correctly matches with its location?

| Tissue | Location |
| --- | --- |
| (1) Smooth muscle | Wall of intestine |
| (2) Areolar tissue | Tendons |
| (3) Transitional epithelium | Tip nose |
| (4) Cuboidal epithelium | Lining of stomach |

Answer (1)

Sol. Smooth muscles are found in the wall of intestine, stomach, and other visceral organs. Smooth muscles are involuntary in nature and are present in all those organs which are not under our conscious control.

Class 11th
Question type: NEET
Difficulty of question: Moderate
Expected time to solve: 35 secs
Topic: Photosynthesis in Higher Plants
Concept: $C_4$ Pathway
Sub-concept: Photorespiration
Concept field: Influencing Factors

53. A plant in your garden avoids photorespiratory losses, has improved water use efficiency, shows high rates of photosynthesis at high temperatures and has improved efficiency of nitrogen utilisation. In which of the following physiological groups would you assign this plant?

(1) $C_3$ (2) $C_4$ (3) CAM (4) Nitrogen fixer

Ans. (2)

Sol. $C_4$ plants have improved water using efficiency, high rate of photosynthesis at high temperatures and nitrogen utilisation efficiency.

Class 12th
Question type: NEET
Difficulty of question: Easy
Expected time to solve: 20 secs
Topic: Evolution
Concept: Evidences for Evolution
Sub-concept: Common Ancestry

Concept field: Homologous Organs

54. Which of the following structures is homologous to the wing of a bird?
(1) Dorsal fin of a Shark
(2) Wing of a Moth
(3) Hind limb of Rabbit
(4) Flipper of Whale

Ans. (4)

Sol. Wings of a bird and flipper of a whale have the same origin but different functions. Hence, they are homologous organs.

Class 11th
Question type: NEET
Difficulty of question: Moderate
Expected time to solve: 35 secs
Topic: Animal Kingdom
Concept: Chordates
Sub-concept: Pisces
Concept field: Chondrichthyes

55. Which of the following characteristic features always holds true for the corresponding group of animals?
(1) Cartilaginous endoskeleton - Chondrichthyes
(2) Viviparous - Mammalia
(3) Possess a mouth with an upper and a lower jaw - Chordata
(4) 3 - chambered heart with one incompletely divided ventricle - Reptilia

Ans. (1)

Sol. Chondrichthyes have cartilaginous endo-skeletons.

Class 12th
Question type: NEET
Difficulty of question: Moderate
Expected time to solve: 30 secs
Topic: Human Health and Diseases
Concept: Cancer
Sub-concept: Causes of Cancer
Concept field: Mutation

56. Which of the following statements is not true for cancer cells in relation to mutations?
(1) Mutations in proto-oncogenes accelerate the cell cycle.
(2) Mutations destroy telomerase inhibitors.
(3) Mutations inactive the cell control.
(4) Mutations inhibit production of telomerase.

Ans. (4)

Sol. Telomerase is required during DNA replication. Cancer cells undergo repeated DNA replication so the telomerase production cannot be inhibited.

Class 11th
Question type: NEET
Difficulty of question: Moderate
Expected time to solve: 30 secs
Topic: Chemical Control and Coordination
Concept: Hormones
Sub-concept: Precursors
Concept field: Amino Acid Derived Hormones

57. The amino acid Tryptophan is the precursor for the synthesis of
(1) Melatonin and Serotonin
(2) Thyroxine and Triiodothyronine
(3) Estrogen and Progesterone
(4) Cortisol and Cortisone

Ans. (1)

Sol. Tryptophan is one of the essential amino acids and the precursor of melatonin and serotonin hormones. Melatonin regulates circadian rhythms and serotonin stabilizes mood, feelings like happiness and well being.

Class 12th
Question type: NEET
Difficulty of question: Moderate
Expected time to solve: 40 secs
Topic: Evolution
Concept: Origin of Life
Sub-concept: Evolution of Life Forms
Concept field: Biogenesis

58. Following are the two statements regarding the origin of life.
(a) The earliest organisms that appeared on the earth were non-green and presumably anaerobes.
(b) The first autotrophic organisms were the chemoautotrophs that never released oxygen.

Of the above statements, which one of the following options is correct?
(1) (a) is correct but (b) is false.
(2) (b) is correct but (a) is false.
(3) Both (a) and (b) are correct.
(4) Both (a) and (b) are false.

Ans. (3)

Sol. Coacervates are the protobionts which are believed to give rise to earliest organisms that were non-photosynthetic and anaerobic. Eventually bacteria like organisms which were chemoautotrophs came into existence. Chemoautotrophs did not use water for fetching hydrogen like green bacteria so never released oxygen.

Class 11th
Question type: NEET
Difficulty of question: Moderate
Expected time to solve: 40 secs
Topic: Breathing and Exchange of Gases
Concept: Transport of Gases
Sub-concept: Transport of Oxygen
Concept field: Influencing Factors

59. Reduction in pH of blood will
(1) reduce the rate of heartbeat.
(2) reduce the blood supply to the brain.
(3) decrease the affinity of haemoglobin with oxygen.
(4) release bicarbonate ions by the liver.

Ans. (3)

Sol. Haemoglobin is the respiratory pigment in the human body which has the affinity to bind with oxygen and carbon dioxide. Haemoglobin's oxygen binding affinity is inversely related to both pH of blood and concentration of carbon dioxide. Carbon dioxide reacts with water to form carbonic acid and an increase in $CO_2$ level in the blood results in decreased pH of blood. This triggers the haemoglobin in releasing its load of oxygen.

Class 12th
Question type: NEET
Difficulty of question: Easy
Expected time to solve: 20 secs
Topic: Evolution
Concept: Evidences
Sub-concept: Anatomical Evidences
Concept field: Convergent Evolution

60. Analogous structures are a result of
(1) Divergent evolution
(2) Convergent evolution
(3) Shared ancestry
(4) Stabilizing selection

Ans. (2)

Sol. Structures that are morphologically similar but not anatomically are called analogous structures. Such a type of structure helps diverse life forms to thrive in a common habitat. Thus, analogy supports convergent evolution.

Class 12th
Question type: NEET
Difficulty of question: Easy
Expected time to solve: 20 secs
Topic: Biotechnology – Principles and Processes
Concept: Tools of rDNA Technology
Sub-concept: Restriction Enzymes
Concept field: First Restriction Enzyme

61. Which of the following is a restriction endonuclease?
(1) *Hind* II
(2) Protease
(3) DNase I
(4) RNase

Ans. (1)

Sol. The first restriction endonuclease is *Hind* II and it always cuts DNA molecules at a specific sequence of six base pairs. This specific sequence is known as the recognition sequence.

Class 12th
Question type: NEET
Difficulty of question: Easy
Expected time to solve: 15 secs
Topic: Ecology
Concept: Ecosystem
Sub-concept: Structure of Ecosystem
Concept field: Contribution of A.G. Tansley

62. The term ecosystem was coined by
(1) E.P. Odum
(2) A.G. Tansley
(3) E. Haeckel
(4) E. Warming

Ans. (2)

Sol. Sir Arthur George Tansley, a pioneer in the science of ecology coined the term ecosystem.

Class 11th
Question type: NEET
Difficulty of question: Easy
Expected time to solve: 20 secs
Topic: Biomolecules

Concept: Biomolecules and Biomacromolecules
Sub-concept: Composition
Concept field: Composition of Biomolecules

63. Which one of the following statements is wrong?
(1) Sucrose is a disaccharide.
(2) Cellulose is a polysaccharide.
(3) Uracil is a pyrimidine.
(4) Glycine is a sulphur containing amino acid.

Ans. (4)

Sol. Glycine is a neutral amino acid. It has two hydrogen atoms attached to the alpha carbon besides one carboxylic group and one amino group.

Class 11th
Question type: NEET
Difficulty of question: Easy
Expected time to solve: 20 secs
Topic: Plant Kingdom
Concept: Cryptogames
Sub-concept: Bryophytes and Pteridophytes
Concept field: Characteristic features

64. In bryophytes and pteridophytes, transport of male gametes requires
(1) Wind (2) Insects (3) Birds (4) Water

Ans. (4)

Sol. Bryophytes and pteridophytes produce flagellated male gametes and they need water to reach the archegonium.

Class 12th
Question type: NEET
Difficulty of question: Moderate
Expected time to solve: 30 secs
Topic: Organisms and Populations
Concept: Populations
Sub-concept: Population Growth
Concept field: Growth Model

65. When does the growth rate of a population following the logistic model equal zero? The logistic model is given as $dN/dt = rN (1–N/K)$:
(1) when N/K is exactly one.
(2) when N nears the carrying capacity of the habitat.
(3) when N/K equals zero.
(4) when the death rate is greater than the birth rate.

Ans. (1)

Sol. In logistic model of population growth
N = Population density at time t
K = Carrying capacity
In the equation, $dN/dt = rN (1–N/K)$ if K is much larger than N, the population can grow quickly; if K is smaller than N, the population will shrink. If K = N, the population will remain the same over time.

Class 12th
Question type: NEET

Difficulty of question: Easy
Expected time to solve: 30 secs
Topic: Reproduction in Flowering Plants
Concept: Flower
Sub-concept: Stamen
Concept field: Pollen Grain

66. Which one of the following statements is not true?
(1) Tapetum helps in the dehiscence of anther
(2) Exine of pollen grains is made up of sporopollenin
(3) Pollen grains of many species cause severe allergies
(4) Stored pollen in liquid nitrogen can be used in the crop breeding programmes

Ans. (1)

Sol. Tapetum nourishes the developing pollen grains.

Class 12th
Question type: NEET
Difficulty of question: Easy
Expected time to solve: 20 secs
Topic: Ecosystem
Concept: Ecological Succession
Sub-concept: Succession of Plants
Concept field: Xerarch Succession

67. Which of the following would appear as the pioneer organisms on bare rocks?
(1) Lichens (2) Liverworts (3) Mosses (4) Green algae

Ans. (1)

Sol. The species that invade any bare, lifeless habitat is called a pioneer species. In xerarch succession lichens are the pioneer organisms as these secrete enzymes that dissolve the rocks and pay way for the other plants to colonise the habitat.

Class 12th
Question type: NEET
Difficulty of question: Easy
Expected time to solve: 20 secs
Topic: Molecular Basis of Inheritance
Concept: Translation
Sub-concept: Initiation
Concept field: Start Codon

68. Which one of the following is the starter codon?
(1) AUG (2) UGA (3) UAA (4) UAG

Ans. (1)

Sol. The translation unit in mRNA is a sequence of RNA that has AUG (start codon) at its 5' end and some additional UTRs for efficient translation process and in the same way at the 3'end stop codon is present, after its UTRs.

Class 11th
Question type: NEET
Difficulty of question: Easy
Expected time to solve: 20 secs
Topic: Animal Kingdom
Concept: Chordates

Sub-concept: Vertebrates
Concept field: Characteristics

69. Which one of the following characteristics is not shared by birds and mammals?
(1) Ossified endoskeleton
(2) Breathing using lungs
(3) Viviparity
(4) Warm blooded nature

Ans. (3)

Sol. Birds are oviparous while mammals are viviparous.

Class 11th
Question type: NEET
Difficulty of question: Easy
Expected time to solve: 30 secs
Topic: The Living World
Concept: Taxonomy
Sub-concept: Scientific Nomenclature
Concept field: Rules

70. Nomenclature is governed by certain universal rules. Which one of the following is contrary to the rules of nomenclature?
(1) Biological names can be written in any language
(2) The first word in a biological name represents the genus name, and the second is a specific epithet
(3) The names are written in Latin and are italicised
(4) When written by hand, the names are to be underlined

Ans. (1)

Sol. Biological names are written in Latin. It is one of the rules according to Binomial Nomenclature which is widely accepted by the biologists.

Class 11th
Question type: NEET
Difficulty of question: Easy
Expected time to solve: 20 secs
Topic: Circulation of Body Fluids
Concept: Cardiac Cycle
Sub-concept: Blood Vessels
Concept field: Blood Pressure

71. Blood pressure in the pulmonary artery is
(1) same as that in the aorta.
(2) more than that in the carotid.
(3) more than that in the pulmonary vein.
(4) less than that in the vena cavae.

Ans. (3)

Sol. In veins the blood always flows under low blood pressure in comparison to arteries.

Class 11th
Question type: NEET
Difficulty of question: Easy
Expected time to solve: 20 secs
Topic: Morphology of Flowering Plants
Concept: Structure of a Seed
Sub-concept: Monocotyledonous Seed

Concept field: Parts

72. Cotyledon of maize grain is called
(1) plumule (2) coleorhiza (3) coleoptile (4) scutellum

Ans. (4)

Sol. In monocot seed there is only one cotyledon called scutellum.

Class 11th
Question type: NEET
Expected time to solve: 20 secs
Topic: Digestion and Absorption
Concept: Human Digestive System
Sub-concept: Digestive Glands
Concept field: Gastric Glands

73. In the stomach, gastric acid is secreted by the
(1) gastrin secreting cells (2) parietal cells
(3) peptic cells (4) acidic cells

Ans. (2)

Sol. In the gastric mucosa numerous gastric glands are present among them parietal cells secrete mucus.

Class 12th
Question type: NEET
Difficulty of question: Easy
Expected time to solve: 20 secs
Topic: Environmental Issues
Concept: Air Pollution
Sub-concept: Ozone Depletion
Concept field: Effects

74. Depletion of which gas in the atmosphere can lead to an increased incidence of skin cancers
(1) Nitrous oxide (2) Ozone (3) Ammonia (4) Methane

Ans. (2)

Sol. UV radiations of wavelengths shorter than UV-B are absorbed by the Earth's Atmosphere considering that the ozone layer is intact. But if the ozone layer gets depleted then the UV-B radiation damages the skin cells and various types of skin cancers can occur.

Class 11th
Question type: NEET
Difficulty of question: Easy
Expected time to solve: 20 secs
Topic: Biological Classification
Concept: Five Kingdom Classification
Sub-concept: Protists
Concept field: Types of Protists

75. Chrysophytes, Euglenoids, Dinoflagellates and Slime moulds are included in the kingdom
(1) Monera (2) Protista (3) Fungi (4) Animalia

Ans. (2)

Sol. Chrysophytes include golden algae (desmids) and diatoms.

Euglenoids include Euglena like flagellates.
Dinoflagellates include biflagellate, photosynthetic forms which are mostly marine. Slime moulds include saprophytes.

Class 11th
Question type: NEET
Difficulty of question: Easy
Expected time to solve: 50 secs
Topic: Photosynthesis
Concept: Role of Stomata
Sub-concept: Gaseous Exchange
Concept field: Mechanism

76. Water vapour comes out from the plant leaf through the stomatal opening. Through the same stomatal opening carbon dioxide diffuses into the plant during photosynthesis. Reason out the above statements using one of the following options.
(1) Both processes cannot happen simultaneously.
(2) Both processes can happen together because the diffusion coefficient of water and $CO_2$ is different.
(3) The above processes happen only during night-time.
(4) One process occurs during daytime, and the other at night.

Ans. (2)

Sol. Gaseous exchange and transpiration occur simultaneously. Diffusion coefficient of carbon dioxide is less in comparison to water vapour so both the gases can diffuse simultaneously.

Class 11th
Question type: NEET
Difficulty of question: Moderate
Expected time to solve: 30 secs
Topic: Excretion and Osmoregulation
Concept: Role of Liver
Sub-concept: Urea Formation
Concept field: Transportation of Urea

77. In mammals, which blood vessel would normally carry the largest amount of urea?
(1) Renal Vein (2) Dorsal Aorta (3) Hepatic Vein (4) Hepatic Portal Vein

Ans. (3)

Sol. The liver is the primary site for the breakdown of excess amino acids into urea. Hence, the hepatic vein would normally carry more urea.

Class 12th
Question type: NEET
Difficulty of question: Easy
Expected time to solve: 20 secs
Topic: Reproduction in Flowering Plants
Concept: Seed Formation
Sub-concept: Apomixis
Concept field: Asexual Reproduction

78. Seed formation without fertilization in flowering plants involves the process of
(1) Sporulation (2) Budding
(3) Somatic hybridization (4) Apomixis

Ans. (4)
Sol. In few flowering plants the egg cell is developed without reduction division so is diploid and it develops into an embryo without fertilization. Such seeds are diploid and are called apomictic seeds. So apomixis is a form of asexual reproduction.

Class 12th
Question type: NEET
Difficulty of question: Easy
Expected time to solve: 20 secs
Topic: Microbes in Human Welfare
Concept: Microbes in Industrial Products
Sub-concept: Chemical, Enzymes and Other Bioactive Molecules
Concept field: Applications

79. Which of the following is wrongly matched in the given table?

| Microbe Product | Application |
|---|---|
| (1) *Trichoderma polysporum* | Cyclosporine A ,immunosuppressive drug |
| (2) *Monascus purpureus* Statins | Lowering of blood cholesterol |
| (3) *Streptococcus Streptokinase* | Removal of clot from blood vessel |
| (4) *Clostridium butylicum* | Lipase Removal of oil stains |

Ans. (4)
Sol. *Clostridium butylicum* is a bacterium used to obtain butyric acid.

Class 12th
Question type: NEET
Difficulty of question: Easy
Expected time to solve: 40 secs
Topic: Principles of Inheritance and Variations
Concept: Inheritance of Two Genes
Sub-concept: Dihybrid Cross
Concept field: Linkage and Recombination

80. In a testcross involving $F_1$ dihybrid flies, more parental-type offspring were produced than the recombinant-type offspring. This indicates.
(1) The two genes are located on two different chromosomes.
(2) Chromosomes failed to separate during meiosis.
(3) The two genes are linked and present on the same chromosome.
(4) Both of the characters are controlled by more than one gene.

Ans. (3)
Sol. Genes are present on the chromosomes in a single linear row. Distance between genes varies and when the distance between two or more genes is less then they tend to move together during gamete formation and do not separate even during crossing over. This phenomenon is called linkage which reduces the chances of recombination and more parental types of offspring are produced.

Class 12th
Question type: NEET
Difficulty of question: Easy
Expected time to solve: 20 secs
Topic: Organisms and Populations
Concept: Life History Variations
Sub-concept: Darwinian Fitness

Concept field: Darwinian Fitness

81. It is much easier for a small animal to run uphill than for a large animal, because
(1) It is easier to carry a small body weight.
(2) Smaller animals have a higher metabolic rate.
(3) Small animals have a lower $O_2$ requirement.
(4) The efficiency of muscles in large animals is less than in small animals.

Ans. (2)

Sol. For faster movement lots of energy is required and to achieve that amount of energy the metabolic rate needs to be faster. That is why smaller animals can easily run uphill.

Class 11th
Question type: NEET
Difficulty of question: Easy
Expected time to solve: 20 secs
Topic: Cell Cycle and Cell Division
Concept: Mitosis Cell Division
Sub-concept: Process of Mitosis
Concept field: Characteristics

82. Which of the following is not a characteristic feature during mitosis in somatic cells?
(1) Spindle fibres
(2) Disappearance of nucleolus
(3) Chromosome movement
(4) Synapsis

Ans. (4)

Sol. Synapsis is a feature that occurs during meiosis cell division.

Class 12th
Question type: NEET
Difficulty of question: Easy
Expected time to solve: 40 secs
Topic: Reproduction in Flowering Plants
Concept: Stamen
Sub-concept: Pollination
Concept field: Pollen Germination

83. Which of the following statements is not correct?
(1) Pollen grains of many species can germinate on the stigma of a flower, but only one pollen tube of the same species grows into the style.
(2) Insects that consume pollen or nectar without bringing about pollination are called pollen/nectar robbers.
(3) Pollen germination and pollen tube growth are regulated by chemical components of pollen interacting with those of the pistil.
(4) Some reptiles have also been reported as pollinators in some plant species.

Ans. (1)

Sol. Pollen germination is species specific and all the pollen grains which belong to that species can germinate on the stigma but the pollen tube which grows faster and reaches the micropyle first is able to release the male gametes into the female gametophyte.

Class 11th
Question type: NEET
Difficulty of question: Easy

Expected time to solve: 20 secs
Topic: Anatomy of Flowering Plants
Concept: Monocot Leaf
Sub-concept: Epidermal Tissue
Concept field: Stomata

84. Specialised epidermal cells surrounding the guard cells are called
(1) Complementary cells (2) Subsidiary cells
(3) Bulliform cells (4) Lenticels

Ans. (2)

Sol. The stomata are surrounded usually by two guard cells which are further surrounded by subsidiary cells. All these cells help in the functioning of stomata.

Class 11th
Question type: NEET
Difficulty of question: Easy
Expected time to solve: 20 secs
Topic: Digestion and Absorption
Concept: Digestive Tract
Sub-concept: Digestive Glands
Concept field: Liver and Pancreas

85. Which of the following guards the opening of the hepato-pancreatic duct into the duodenum?
(1) Semilunar valve (2) Ileocaecal valve
(3) Pyloric sphincter (4) Sphincter of Oddi

Ans. (4)

Sol. The bile duct and the pancreatic duct open together into the duodenum as the common hepato-pancreatic duct which is guarded by a sphincter called sphincter of Oddi.

Class 11th
Question type: NEET
Difficulty of question: Easy
Expected time to solve: 20 secs
Topic: Morphology of Flowering Plants
Concept: Shoot System
Sub-concept: Stem
Concept field: Modifications of Stem

86. Stems modified into flat green organs performing the functions of leaves are known as
(1) Cladodes (2) Phyllodes (3) Phylloclades (4) Scales

Ans. (3)

Sol. In cactus the leaves are converted into spines to reduce transpiration, so the stem becomes green, flat, leaf-like and performs photosynthesis. Such a stem is called phylloclade.

Class 11th
Question type: NEET
Difficulty of question: Easy
Expected time to solve: 20 secs
Topic: Biological Classification
Concept: Kingdom Monera

Sub-concept: Archaebacteria
Concept field: Methanogens

87. The primitive prokaryotes responsible for the production of biogas from the dung of ruminant animals, include the
(1) Halophiles
(2) Thermoacidophiles
(3) Methanogens
(4) Eubacteria

Ans. (3)

Sol. Methanogens are archaebacteria which are specialised to survive in most harsh habitats. These methanogens are used to produce methane, a component of biogas in biogas plants.

Class 12th
Question type: NEET
Difficulty of question: Easy
Expected time to solve: 20 secs
Topic: Environmental Issues
Concept: Domestic Sewage and Industrial Effluents
Sub-concept: Sewage Discharge in Rivers
Concept field: Effect

88. A river with an inflow of domestic sewage rich inorganic waste may result in
(1) Drying of the river very soon due to algal bloom.
(2) Increased population of aquatic food web organisms.
(3) An increased production of fish due to biodegradable nutrients.
(4) Death of fish due to lack of oxygen.

Ans. (4)

Sol. At the site of sewage discharge into a river the amount of dissolved oxygen is reduced tremendously due to rise in biochemical oxygen demand, this results in the death of fish and other aquatic animals.

Class 12th
Question type: NEET
Difficulty of question: Moderate
Expected time to solve: 30 secs
Topic: Principles of Inheritance and Variations
Concept: Chromosomal Disorders
Sub-concept: Polyploidy
Concept field: Nondisjunction Mechanism

89. A cell at the telophase stage is observed by a student in a plant brought from the field. He tells his teacher that this cell is not like other cells at telophase stage. There is no formation of cell plates and thus the cell contains a greater number of chromosomes as compared to other dividing cells. This would result in
(1) Aneuploidy
(2) Polyploidy
(3) Somaclonal variation
(4) Polyteny

Ans. (2)

Sol. Polyploidy means increase in a whole set of chromosomes i.e., Triticum aestivum is an allohexaploid wheat variety. Polyploidy is often seen in plants.

Class 11th
Question type: NEET
Difficulty of question: Easy
Expected time to solve: 20 secs

Topic: Biomolecules
Concept:Lipids
Sub-concept: Fats
Concept field: Chemical Structure

90. A typical fat molecule is made up of
(1) Three glycerol molecules and one fatty acid molecule
(2) One glycerol and three fatty acid molecules
(3) One glycerol and one fatty acid molecule
(4) Three glycerol and three fatty acid molecules

Ans. (2)

Sol. In a typical fat molecule three fatty acids are found esterified with glycerol.